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CS 215: Task 01 1: Due: 08-Aug Time: as mentioned in Piazza. No extensions.

- Your answers should be in your handwriting. Try to be neat and clear. If an LLM can read the data correctly, we might be able to.
- Submission Guidelines. See `piazza` for standard static instructions.
- Your submission folder after extraction should look something like:

```
23B0019_task01/
├── readme.txt
└── task01.pdf
```

  You should probably name your zip file correctly. Notice how, AFTER extraction, the two files are inside the folder containing the roll number.

---

0. Please write the honor code in your handwriting, with the understanding that it is true.

1. A red die, a blue die, and a yellow die (all fair six-sided dice) are rolled. Let B, R, and Y denote the numbers on the blue, red and yellow dice respectively.

   We are interested in the event $B < Y < R$.

   (a) What is the probability that no two dice land on the same number?

   (b) Given that no two dice land on the same number, what is the conditional probability that $B < Y < R$?

(c) Find $P(B < Y < R)$ using part (a).

(d) If we regard the outcome as the ordered vector (B, R, Y), how many outcomes are there in the sample space?

(e) Without using the answer to part (c), can you determine the number of outcomes satisfying $B < Y < R$ in an alternate way?

(f) Use the results of parts (d) and (e) to verify the answer to part (c).

2. The sample mean and sample variance of five data values are, respectively,

$$\bar{x} = 104, \qquad s^2 = 16.$$

If three of the data values are $102, 100$, and 106, determine the other two data values.

3. A computationally efficient way to compute the sample mean and sample variance is given by

$$\bar{x}_{j+1} = \bar{x}_j + \frac{x_{j+1} - \bar{x}_j}{j+1},$$

and

$$s^2_{j+1} = \left(1 - \frac{1}{j}\right) s^2_j + (j+1)(\bar{x}_{j+1} - \bar{x}_j)^2.$$

(a) Use these formulas to compute the sample mean and sample variance of

$$3,\ 4,\ 7,\ 2,\ 9,\ 6.$$

(b) Verify your results in part (a) using the usual formulas for sample mean and sample variance.

(c) Verify the recursive formula

$$\bar{x}_{j+1} = \bar{x}_j + \frac{x_{j+1} - \bar{x}_j}{j+1}.$$

4. An engineering system consisting of n independent components is said to be a k-out-of-n system if it functions whenever at least k components function.

   (a) If the ith component functions with probability P_i, $i = 1, 2, 3, 4$, compute the probability that a 2-out-of-4 system functions.

   (b) Generalize your reasoning from part (a) to write a compact expression for the probability that a 3-out-of-5 system functions. Use summation or subset notation; do not list all outcomes individually.

CS 215: Task 01 1: Due: 08-Aug Time: as mentioned in Piazza. No extensions.

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- Submission Guidelines. See `piazza` for standard static instructions.
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├── readme.txt
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You should probably name your zip file correctly. Notice how, AFTER extraction, the two files are inside the folder containing the roll number.

---

0. Please write the honor code in your handwriting, with the understanding that it is true.

1. A red die, a blue die, and a yellow die (all fair six-sided dice) are rolled. Let B, R, and Y denote the numbers on the blue, red and yellow dice respectively.

   We are interested in the event $B < Y < R$.

   (a) What is the probability that no two dice land on the same number?
   Answer:
   There are $6^3 = 216$ equally likely outcomes.
   For all three numbers to be distinct,

   $$6 \times 5 \times 4 = 120$$

   ordered outcomes are possible.
   Thus,

   $$P(\text{all distinct}) = \frac{120}{216} = \frac{5}{9}.$$

   Marking scheme: 3 marks.

   (b) Given that no two dice land on the same number, what is the conditional probability that $B < Y < R$?
   Answer:
   Conditioned on distinct values, the six possible orderings of the three dice are equally likely:

   $$BYR,\ BRY,\ YBR,\ YRB,\ RBY,\ RYB.$$

   Only one ordering satisfies

   $$B < Y < R.$$

   Hence,

$$P(B < Y < R \mid \text{distinct}) = \frac{1}{6}.$$

Marking scheme: 3 marks.

(c) Find $P(B < Y < R)$ using part (a).

Answer:

Using part (a),

$$P(B < Y < R) = P(B < Y < R \mid \text{distinct})P(\text{distinct}).$$

Therefore,

$$P(B < Y < R) = \frac{1}{6} \cdot \frac{5}{9} = \boxed{\frac{5}{54}}.$$

Marking scheme: 3 marks.

(d) If we regard the outcome as the ordered vector (B, R, Y), how many outcomes are there in the sample space?

Answer:

Each die has six possible outcomes.

Hence,

$$6 \times 6 \times 6 = 216.$$

Marking scheme: 2 marks.

(e) Without using the answer to part (c), can you determine the number of outcomes satisfying $B < Y < R$ in an alternate way?

Answer:

Choose any three distinct numbers from $\{1, \ldots, 6\}$.

The number of choices is

$$\binom{6}{3} = 20.$$

For each choice, exactly one assignment satisfies

$$B < Y < R.$$

Hence,

$$20$$

outcomes satisfy the required ordering.

Marking scheme: 4 marks.

(f) Use the results of parts (d) and (e) to verify the answer to part (c).

Answer:

Using parts (d) and (e),

$$P(B < Y < R) = \frac{20}{216} = \frac{5}{54},$$

which agrees with the answer obtained in part (c).
Marking scheme: 2 marks.

2. The sample mean and sample variance of five data values are, respectively,

$$\bar{x} = 104, \qquad s^2 = 16.$$

If three of the data values are $102, 100$, and 106, determine the other two data values.

Answer:

Let the unknown observations be x and y.

Since the sample mean is 104,

$$\frac{102 + 100 + 106 + x + y}{5} = 104,$$

which gives

$$x + y = 212.$$

Since the sample variance is

$$s^2 = \frac{\sum(x_i - \bar{x})^2}{n - 1} = 16,$$

we have

$$\sum(x_i - 104)^2 = 4(16) = 64.$$

Hence,

$$(x - 104)^2 + (y - 104)^2 = 64 - 24 = 40.$$

Solving the above equations, we get the following solutions

Thus,

$$(x, y) = (110, 102)$$

or

$$(x, y) = (102, 110).$$

Therefore, the two missing observations are

$$\boxed{102 \text{ and } 110.}$$

Marking scheme (6 marks):

- Mean equation: 2 marks
- Variance equation: 2 marks
- Solving quadratic: 1 mark
- Correct answer: 1 mark

3. A computationally efficient way to compute the sample mean and sample variance is given by

$$\bar{x}_{j+1} = \bar{x}_j + \frac{x_{j+1} - \bar{x}_j}{j+1},$$

and

$$s^2_{j+1} = \left(1 - \frac{1}{j}\right) s^2_j + (j+1)(\bar{x}_{j+1} - \bar{x}_j)^2.$$

(a) Use these formulas to compute the sample mean and sample variance of

$$3,\ 4,\ 7,\ 2,\ 9,\ 6.$$

Answer:
The computations are summarized below.

j	$\bar{x}_j$	s^2_j
1	3	0
2	3.5	0.5
3	$\frac{14}{3}$	$\frac{13}{3}$
4	4	$\frac{14}{3}$
5	5	8.5
6	$\frac{31}{6}$	$\frac{86}{15}$

Hence,

$$\boxed{\bar{x} = \frac{31}{6}, \qquad s^2 = \frac{86}{15}.}$$

Marking scheme (6 marks):

- Correct recursive means: 3 marks
- Correct recursive variances: 2 marks
- Final answer: 1 mark

(b) Verify your results in part (a) using the usual formulas for sample mean and sample variance.
Answer:
The sample mean is

$$\bar{x} = \frac{3+4+7+2+9+6}{6} = \frac{31}{6}.$$

Next,

$$\sum (x_i - \bar{x})^2 = \frac{172}{6} = \frac{86}{3}.$$

Therefore,

$$s^2 = \frac{1}{6-1}\left(\frac{86}{3}\right) = \frac{86}{15}.$$

Thus the values agree with those obtained in part (a).
Marking scheme (4 marks):

- Mean computation: 1 mark
- Variance computation: 2 marks
- Verification: 1 mark

(c) Verify the recursive formula

$$\bar{x}_{j+1} = \bar{x}_j + \frac{x_{j+1} - \bar{x}_j}{j+1}.$$

Answer:
By definition,

$$\bar{x}_{j+1} = \frac{x_1 + \cdots + x_j + x_{j+1}}{j+1}.$$

Since

$$x_1 + \cdots + x_j = j\bar{x}_j,$$

we obtain

$$\bar{x}_{j+1} = \frac{j\bar{x}_j + x_{j+1}}{j+1}.$$

Rearranging,

$$\bar{x}_{j+1} = \bar{x}_j + \frac{x_{j+1} - \bar{x}_j}{j+1},$$

which proves the desired identity.
Marking scheme (4 marks):

- Start from definition: 2 marks
- Algebraic manipulation: 1 mark
- Final identity: 1 mark

4. An engineering system consisting of n independent components is said to be a k-out-of-n system if it functions whenever at least k components function.

(a) If the ith component functions with probability P_i, $i = 1, 2, 3, 4$, compute the probability that a 2-out-of-4 system functions.

Answer:

The system functions if at least two of the four components function. Thus we consider the cases of exactly two, exactly three, and exactly four functioning components.

Exactly two components function:

$$\sum_{i=1}^{3} \sum_{j=i+1}^{4} P_i P_j \prod_{\substack{k=1 \\ k \neq i,j}}^{4} (1 - P_k).$$

Exactly three components function:

$$\sum_{i=1}^{4}(1-P_i)\prod_{\substack{j=1\\ j\neq i}}^{4}P_j.$$

All four components function:

$$\prod_{i=1}^{4}P_i.$$

Hence,

$$P(\text{system functions}) = \sum_{i=1}^{3}\sum_{j=i+1}^{4}P_iP_j\prod_{\substack{k=1\\ k\neq i,j}}^{4}(1-P_k) + \sum_{i=1}^{4}(1-P_i)\prod_{\substack{j=1\\ j\neq i}}^{4}P_j + \prod_{i=1}^{4}P_i.$$

Marking scheme (6 marks):

- Exactly two components: 3 marks
- Exactly three components: 2 marks
- All four components: 1 mark

(b) Generalize your reasoning from part (a) to write a compact expression for the probability that a 3-out-of-5 system functions. Use summation or subset notation; do not list all outcomes individually.

Answer:

The system functions if at least three of the five components function. Hence we consider the cases of exactly three, exactly four, and exactly five functioning components.

Exactly three components function:

$$\sum_{i=1}^{3}\sum_{j=i+1}^{4}\sum_{k=j+1}^{5}P_iP_jP_k\prod_{\substack{\ell=1\\ \ell\neq i,j,k}}^{5}(1-P_\ell).$$

Exactly four components function:

$$\sum_{i=1}^{5}(1-P_i)\prod_{\substack{j=1\\ j\neq i}}^{5}P_j.$$

All five components function:

$$\prod_{i=1}^{5}P_i.$$

Therefore,

$$P(\text{system functions}) = \sum_{i=1}^{3}\sum_{j=i+1}^{4}\sum_{k=j+1}^{5}P_iP_jP_k\prod_{\substack{\ell=1\\ \ell\neq i,j,k}}^{5}(1-P_\ell) + \sum_{i=1}^{5}(1-P_i)\prod_{\substack{j=1\\ j\neq i}}^{5}P_j + \prod_{i=1}^{5}P_i.$$

Marking scheme (6 marks):

- Exactly three components: 3 marks
- Exactly four components: 2 marks
- All five components: 1 mark

CS 215: Task 02 Due: 15-Aug Time: as mentioned in Piazza. No extensions.

- Your answers should be in your handwriting. Try to be neat and clear. If an LLM can read the data correctly, we might be able to.
- You are highly encouraged to NOT use an LLM to solve this worksheet. If you get stuck, and want to use an LLM, make sure you tell the LLM that you don't want a solution, but an answer to your question. Don't say "give me a hint". Ask: "I am trying this problem in this way, am I right? Answer yes or no".
- The problems build upon concepts, so it is recommended you approach them linearly. It is recommended that you do not attempt the questions out of order.
- For some problems, check out the hints carefully.
- For brevity, we will use probability density function as density function or p.d.f., or simply pdf.
- In all cases, when you write an answer, give reasons. The space available is indicate of the length of the reason desired.

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0. Please write the honor code in your handwriting, with the understanding that it is true.

1. X is a continuous random variable with pdf

$$f(x) = \begin{cases} C(4x - 2x^2), & 0 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

   (a) C is

   (b) $\mathbb{P}\{X > 1\}$ is

2. The amount of time, in hours, that a computer functions before breaking down is a continuous random variable with pdf

$$f(x) = \begin{cases} \lambda e^{-x/100}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

(a) What is the probability that a computer will function between 50 and 150 hours before breaking down?

(b) What is the probability that it will function less than 100 hours?

3. The joint pdf of X and Y is given by

$$f(x, y) = \frac{6}{7}\left(x^2 + \frac{xy}{2}\right), \qquad 0 < x < 1, \quad 0 < y < 2.$$

(a) Verify that this is indeed a joint density function.

(b) Compute the density function of X.

(c) $\mathbb{P}\{X > Y\}$ is

4. X and Y are independent continuous random variables. Show that

(a)
$$P\{X + Y \leq a\} = \int_{-\infty}^{\infty} F_X(a - y) f_Y(y)\, dy$$
where f_Y is the density function of Y, and F_X is the distribution function of X.

(b)
$$P\{X \leq Y\} = \int_{-\infty}^{\infty} F_X(y) f_Y(y)\, dy$$
where f_Y is the density function of Y, and F_X is the distribution function of X.

5. In a certain town, there exist 100 buses out of which 1 is red and 99 are blue. XYZ observes a serious accident caused by a bus at night and remembers that the bus was red in color. Hence, the police arrest the driver of the red bus. The driver pleads innocence. Now, a benevolent lawyer decides to defend the distressed bus driver in court. The lawyer ropes in an ophthalmologist to test XYZ's ability to differentiate between the colors red and blue, under illumination conditions similar to those that existed that fateful night. The ophthalmologist's expert deposition to the court is that XYZ sees red objects as red 99% of the time and blue objects as red 2% of the time.

   If you are the defense lawyer how will you argue? That is, compute the probability that the bus was red, when XYZ thought it to be red, possibly correctly, possibly incorrectly. Provide reasons.

   Teaching Team Notes:

   (a) In problems like this, it is good to start with something like: Let X be the event with such-and-such property. We are not sure about the outcome, so we assign a probability to this event. Similarly build other events, or random variables as the case may be, depending on how you like to think about events and random variables.

   (b) In this particular case, consider conditional probability given the wording of the problem statement.

6. $X_1, X_2, \ldots, X_n$ are independent random variables each having the common (canonical) uniform density function

$$f(x) = \begin{cases} 1, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Teaching Team Notes:

(a) In a problem like this which may seem out of reach, first restrict the variables to n=2 and see if you can reason.

(b) When we need the expectation, we need the pdf. For continuous random variables, consider looking at the distribution function and then differentiating it to get the pdf.

(c) Since the problem specification talks of independent, the hint is to think of the joint pdf and then use the product formula.

$E[\text{Max}(X_1, \ldots, X_n)]$ is

7. A community consists of 100 married couples. If 50 members of the community die, what is the expected number of marriages that remain intact? Assume that the set of people who die is equally likely to be any of the $\binom{200}{50}$ groups of size 50.

Teaching Team Notes:

(a) In problems like this, you are looking for using the linearity of functions along with indicator variables.

(b) For this problem, this indicator variable is relevant

$$X_i = \begin{cases} 1, & \text{if neither member of couple } i \text{ dies,} \\ 0, & \text{otherwise.} \end{cases}$$

(c) Also pay attention to when the marriage is intact, and when a person dies.

CS 215: Task 02 Due: 15-Aug Time: as mentioned in Piazza. No extensions.

0. Please write the honor code in your handwriting, with the understanding that it is true.

1. X is a continuous random variable with pdf

$$f(x) = \begin{cases} C(4x - 2x^2), & 0 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

(a) C is

C is 3/8.

Teaching Team Notes: It is advisable to answer the question :). This implies that, in this case, you state what is C is (either at the end or at the beginning, but state the answer).

Since f is a probability density function, we must have $\int_{-\infty}^{\infty} f(x)\,dx = 1$, which means

$$C \int_0^2 (4x - 2x^2)\,dx = 1$$

$$C \left[2x^2 - \frac{2x^3}{3}\right]_{x=0}^{x=2} = 1$$

$$C = \frac{3}{8}$$

(b) $\mathbb{P}\{X > 1\}$ is

$$P\{X > 1\} = \int_1^{\infty} f(x)\,dx = \frac{3}{8}\int_1^2 (4x - 2x^2)\,dx = \frac{1}{2}$$

2. The amount of time, in hours, that a computer functions before breaking down is a continuous random variable with pdf

$$f(x) = \begin{cases} \lambda e^{-x/100}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

(a) What is the probability that a computer will function between 50 and 150 hours before breaking down?

Since $f(x)$ is a probability density function,

$$\int_{-\infty}^{\infty} f(x)\,dx = 1.$$

Thus,

$$\lambda \int_0^\infty e^{-x/100}\,dx = 1$$

$$\lambda(100) = 1 \quad \Longrightarrow \quad \lambda = \frac{1}{100}.$$

Therefore,

$$P(50 < X < 150) = \int_{50}^{150} \frac{1}{100} e^{-x/100}\,dx$$

$$= \left[-e^{-x/100}\right]_{50}^{150} = e^{-1/2} - e^{-3/2} \approx 0.3834.$$

Therefore answer is $\boxed{e^{-1/2} - e^{-3/2}}$.

(b) What is the probability that it will function less than 100 hours?

$$P(X < 100) = \int_0^{100} \frac{1}{100} e^{-x/100}\,dx$$

$$= \left[-e^{-x/100}\right]_0^{100} = 1 - e^{-1} \approx 0.6321.$$

Therefore answer is $\boxed{1 - e^{-1}}$.

Teaching Team Notes: It is OK to leave the answer without putting the decimal values. $(1 - 1/e)$ is ok, but leaving the answer as $(-e^{-1} + 1)$ is awkward. The rule of thumb in the exam is try to make the answer easy to read, so a box around the answer is recommended. In homeworks, though, it is upto you, and we may not follow the same in these answers.

3. The joint pdf of X and Y is given by

$$f(x, y) = \frac{6}{7}\left(x^2 + \frac{xy}{2}\right), \qquad 0 < x < 1, \quad 0 < y < 2.$$

(a) Verify that this is indeed a joint density function.
We need to verify that

$$\int_0^1 \int_0^2 f(x, y)\,dy\,dx = 1.$$

Thus,

$$\int_0^1 \int_0^2 \frac{6}{7}\left(x^2 + \frac{xy}{2}\right) dy\,dx = \frac{6}{7}\int_0^1 \left(2x^2 + x\right) dx$$

$$= \frac{6}{7}\left(\frac{2}{3} + \frac{1}{2}\right) = 1.$$

Since $f(x, y) \geq 0$ on its support, it is a joint density function.

Teaching Team Notes: Note this last line for the exam. You may lose marks if your answer is incomplete.

(b) Compute the density function of X.
The density function of X is

$$\begin{aligned} f_X(x) &= \int_0^2 f(x,y)\,dy \\ &= \frac{6}{7}\int_0^2 \left(x^2 + \frac{xy}{2}\right) dy \\ &= \frac{6}{7}(2x^2 + x), \qquad 0 < x < 1. \end{aligned}$$

Hence,

$$f_X(x) = \begin{cases} \frac{6}{7}(2x^2 + x), & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Teaching Team Notes: Make sure you capture all cases.

(c) $\mathbb{P}\{X > Y\}$ is
The region $X > Y$ is given by $0 < x < 1$ and $0 < y < x$. Therefore,

$$\begin{aligned} \mathbb{P}\{X > Y\} &= \int_0^1 \int_0^x \frac{6}{7}\left(x^2 + \frac{xy}{2}\right) dy\,dx \\ &= \frac{6}{7}\int_0^1 \left(x^3 + \frac{x^3}{4}\right) dx \\ &= \frac{15}{14}\int_0^1 x^3\,dx \\ &= \boxed{\frac{15}{56}}. \end{aligned}$$

4. X and Y are independent continuous random variables. Show that

(a)

$$P\{X + Y \le a\} = \int_{-\infty}^{\infty} F_X(a - y) f_Y(y)\,dy$$

where f_Y is the density function of Y, and F_X is the distribution function of X.
Since X and Y are independent, their joint density is

$$f_{X,Y}(x,y) = f_X(x) f_Y(y).$$

(a) Conditioning on $Y = y$,

$$\begin{aligned} P\{X + Y \le a\} &= \int_{-\infty}^{\infty} P\{X \le a - y \mid Y = y\} f_Y(y)\,dy \\ &= \int_{-\infty}^{\infty} P\{X \le a - y\} f_Y(y)\,dy \\ &= \boxed{\int_{-\infty}^{\infty} F_X(a - y) f_Y(y)\,dy}. \end{aligned}$$

(b)

$$P\{X \le Y\} = \int_{-\infty}^{\infty} F_X(y) f_Y(y)\, dy$$

where f_Y is the density function of Y, and F_X is the distribution function of X.

(b) Similarly, conditioning on $Y = y$,

$$\begin{aligned} P\{X \le Y\} &= \int_{-\infty}^{\infty} P\{X \le y \mid Y = y\} f_Y(y)\, dy \\ &= \int_{-\infty}^{\infty} P\{X \le y\} f_Y(y)\, dy \\ &= \boxed{\int_{-\infty}^{\infty} F_X(y) f_Y(y)\, dy}. \end{aligned}$$

5. In a certain town, there exist 100 buses out of which 1 is red and 99 are blue. XYZ observes a serious accident caused by a bus at night and remembers that the bus was red in color. Hence, the police arrest the driver of the red bus. The driver pleads innocence. Now, a benevolent lawyer decides to defend the distressed bus driver in court. The lawyer ropes in an ophthalmologist to test XYZ's ability to differentiate between the colors red and blue, under illumination conditions similar to those that existed that fateful night. The ophthalmologist's expert deposition to the court is that XYZ sees red objects as red 99% of the time and blue objects as red 2% of the time.

   If you are the defense lawyer how will you argue? That is, compute the probability that the bus was red, when XYZ thought it to be red, possibly correctly, possibly incorrectly. Provide reasons.

   Teaching Team Notes:

   (a) In problems like this, it is good to start with something like: Let X be the event with such-and-such property. We are not sure about the outcome, so we assign a probability to this event. Similarly build other events, or random variables as the case may be, depending on how you like to think about events and random variables.

   (b) In this particular case, consider conditional probability given the wording of the problem statement.

   Solution:

   Let R denote the event that the bus is actually red, and O denote the event that XYZ observes the bus to be red.

   We have

$$P(R) = \frac{1}{100}, \qquad P(R^c) = \frac{99}{100},$$

   and

$$P(O \mid R) = 0.99, \qquad P(O \mid R^c) = 0.02.$$

   By Bayes' theorem,

$$
\begin{aligned}
P(R \mid O) &= \frac{P(O \mid R)P(R)}{P(O \mid R)P(R) + P(O \mid R^c)P(R^c)} \\
&= \frac{0.99(0.01)}{0.99(0.01) + 0.02(0.99)} \\
&= \frac{0.0099}{0.0297} = \boxed{\frac{1}{3}}.
\end{aligned}
$$

Thus, the main argument of the defense lawyer is that even after XYZ observed a red bus, the probability that the bus was actually red is only

$$\boxed{\frac{1}{3} \approx 33.3\%}.$$

Hence, there is a substantial probability that the bus was actually blue.

6. $X_1, X_2, \ldots, X_n$ are independent random variables each having the common (canonical) uniform density function

$$f(x) = \begin{cases} 1, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Teaching Team Notes:

(a) In a problem like this which may seem out of reach, first restrict the variables to n=2 and see if you can reason.

(b) When we need the expectation, we need the pdf. For continuous random variables, consider looking at the distribution function and then differentiating it to get the pdf.

(c) Since the problem specification talks of independent, the hint is to think of the joint pdf and then use the product formula.

$E[\text{Max}(X_1, \ldots, X_n)]$ is

Since $X_1, \ldots, X_n$ are independent and uniformly distributed on $(0, 1)$, let

$$M = \text{Max}(X_1, \ldots, X_n),$$

For the maximum,

$$
\begin{aligned}
F_M(x) &= P(M \le x) \\
&= P(X_1 \le x, \ldots, X_n \le x) \\
&= x^n, \qquad 0 < x < 1.
\end{aligned}
$$

Thus,

$$f_M(x) = nx^{n-1}.$$

Therefore,

$$E[M] = \int_0^1 x\, n x^{n-1}\, dx = n \int_0^1 x^n\, dx = \boxed{\frac{n}{n+1}}.$$

7. A community consists of 100 married couples. If 50 members of the community die, what is the expected number of marriages that remain intact? Assume that the set of people who die is equally likely to be any of the $\binom{200}{50}$ groups of size 50.

Teaching Team Notes:

(a) In problems like this, you are looking for using the linearity of functions along with indicator variables.

(b) For this problem, this indicator variable is relevant

$$X_i = \begin{cases} 1, & \text{if neither member of couple } i \text{ dies,} \\ 0, & \text{otherwise.} \end{cases}$$

(c) Also pay attention to when the marriage is intact, and when a person dies.

Solution:

Let X be the number of marriages that remain intact. Then

$$X = \sum_{i=1}^{100} X_i.$$

By linearity of expectation,

$$E[X] = \sum_{i=1}^{100} E[X_i] = 100 P(X_i = 1).$$

For a particular couple, the probability that neither member dies is

$$P(X_i = 1) = \frac{150}{200} \cdot \frac{149}{199}.$$

Hence,

$$\begin{aligned} E[X] &= 100 \left(\frac{150}{200}\right)\left(\frac{149}{199}\right) \\ &= \frac{11175}{199} \\ &\approx 56.16. \end{aligned}$$

CS 215: Task 03: Due: 22-Aug Time: as mentioned in Piazza

- Please write (only if true) the honor code. If you used any source (person or thing) explicitly state it. This is an individual assignment.
- Your answers should be in your handwriting.
- Problems are not necessarily in order of difficulty.

---

0. Please write the honor code in your handwriting, with the understanding that it is true.

1. A person tried by a 3-judge panel is declared guilty if at least 2 judges cast votes of guilty. Suppose that when the defendant is in fact guilty, each judge will independently vote guilty with probability 0.7, whereas when the defendant is in fact innocent, this probability drops to 0.2. If 70 percent of defendants are guilty, compute (see hints ahead) the conditional probability that judge number 3 votes guilty given that

   (a) judges 1 and 2 vote guilty,

   (b) judges 1 and 2 cast one guilty and one not guilty vote,

   (c) judges 1 and 2 both cast not guilty votes.

Hint: This problem has three elements.

(a) Joint probabilties, the chain rule: P(A,B,C) = P(A — B, C) P(B—C) P(C), and Bayes' rule

(b) The second line with the word "independent" is the heart of the simplification of the joint probabilities

(c) If you get stuck trying to solve this problem, consider changing the problem to a two judge scenario, with the artificial rule that a defendent is guilty if either of the two judges declares the defendant to be guilty.

2. Let X and Y be random variables with finite variances

$$\mathrm{Var}(X) = \sigma_X^2 \qquad \text{and} \qquad \mathrm{Var}(Y) = \sigma_Y^2.$$

(a) As something unrelated to X and Y – but will be used later on in (d) – prove that if $\mathrm{Var}(Z) = 0$, then

$$P\{Z = \mathbb{E}[Z]\} = 1.$$

(b) Using the fact that variance is always nonnegative, show that

$$0 \leq \mathrm{Var}\left(\frac{X}{\sigma_X} + \frac{Y}{\sigma_Y}\right)$$

implies that

$$-1 \leq \mathrm{Corr}(X, Y).$$

(c) Similarly, use

$$0 \leq \text{Var}\left(\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y}\right)$$

to prove that

$$-1 \leq \text{Corr}(X, Y) \leq 1.$$

(d) Using the result from part (a), prove that if

$$\text{Corr}(X, Y) = 1$$

or

$$\text{Corr}(X, Y) = -1,$$

then X and Y must satisfy a linear relationship of the form

$$Y = a + bX,$$

where $b > 0$ when $\text{Corr}(X, Y) = 1$ and $b < 0$ when $\text{Corr}(X, Y) = -1$.

3. Consider n independent trials, each of which results in one of the three outcomes 1, 2, or 3, with corresponding probabilities p_1, p_2, and p_3, where

$$p_1 + p_2 + p_3 = 1.$$

Let N_i denote the number of trials that result in outcome i.

Find

$$\mathbb{E}[N_i]$$

4. Let (X, Y, Z) have joint probability density

$$f_{X,Y,Z}(x, y, z) = \begin{cases} c(x + z), & 0 < x < y < z < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Determine the constant c.

(b) Compute

$$P(X + Y < Z).$$

Hint: Refer to example 4.3c from the textbook

(c) Compute

$$E[XZ].$$

5. Let $X \geq 0$ be a random variable satisfying

$$E[X] = 4, \qquad E[X^2] = 24, \qquad E[X^3] = 192.$$

Use Markov's inequality, applied appropriately to X, X^2, and X^3, to obtain a tight bound from the given information on

$$P(X \geq 10).$$

6. Let $X > 0$ be a random variable satisfying

$$E[X] = \mu.$$

For some constant $a > 0$, define

$$g(x) = x + \frac{a}{x}.$$

(a) Use Jensen's inequality to obtain a bound, in terms of μ and a, for

$$E\left[X + \frac{a}{X}\right].$$

(b) If μ may also be chosen freely, determine the value of μ to minimize the Jensen bound.

7. Let $X_1, \ldots, X_{25}$ be identically distributed random variables with

$$E[X_i] = 100, \qquad \mathrm{Var}(X_i) = 400.$$

Suppose that, for $i \neq j$,

$$\mathrm{Cov}(X_i, X_j) = 40.$$

Let

$$\overline{X} = \frac{1}{25}\sum_{i=1}^{25} X_i.$$

(a) Use Chebyshev's inequality to obtain a lower bound for

$$P(92 < \overline{X} < 108).$$

(b) Compare this bound with the one obtained if the X_i's were independent. Expand on your observations on the the two differing bounds.

CS 215: Task 03: Due: 22-Aug Time: as mentioned in Piazza

- Please write (only if true) the honor code. If you used any source (person or thing) explicitly state it. This is an individual assignment.
- Your answers should be in your handwriting.
- Problems are not necessarily in order of difficulty.

---

0. Please write the honor code in your handwriting, with the understanding that it is true.

1. A person tried by a 3-judge panel is declared guilty if at least 2 judges cast votes of guilty. Suppose that when the defendant is in fact guilty, each judge will independently vote guilty with probability 0.7, whereas when the defendant is in fact innocent, this probability drops to 0.2. If 70 percent of defendants are guilty, compute (see hints ahead) the conditional probability that judge number 3 votes guilty given that

   Hint: This set of problem has three elements.

   (a) Joint probabilties, the chain rule: $P(A, B, C) = P(A|B, C)P(B|C)P(C)$, and Bayes' rule

   (b) The second line with the word "independent" is the heart of the simplification of the joint probabilities

   (c) If you get stuck trying to solve this problem, consider changing the problem to a two judge scenario, with the artificial rule that a defendent is guilty if either of the two judges declares the defendant to be guilty.

   (a) judges 1 and 2 vote guilty, Let G denote that the defendant is guilty and I denote that the defendant is innocent. We want to find

   $$P(J_3 = G \mid J_1 = G, J_2 = G).$$

   Since each judge votes independently given the defendant's status,

   $$P(J_1 = G, J_2 = G \mid G) = P(J_1 = G \mid G) \cdot P(J_2 = G \mid G) = (0.7)^2 = 0.49,$$

   and

   $$P(J_1 = G, J_2 = G \mid I) = P(J_1 = G \mid I) \cdot P(J_2 = G \mid I) = (0.2)^2 = 0.04.$$

   By Bayes' theorem,

   $$P(G \mid J_1 = G, J_2 = G) = \frac{P(J_1 = G, J_2 = G \mid G)P(G)}{P(J_1 = G, J_2 = G \mid G)P(G) + P(J_1 = G, J_2 = G \mid I)P(I)}$$

   $$= \frac{(0.49)(0.7)}{(0.49)(0.7) + (0.04)(0.3)} = \frac{0.343}{0.355} = \frac{343}{355}.$$

   Therefore,

   $$P(I \mid J_1 = G, J_2 = G) = 1 - \frac{343}{355} = \frac{12}{355}.$$

Now, using the conditional independence of judge 3's vote,

$$\begin{aligned} P(J_3 = G \mid J_1 = G, J_2 = G) &= P(J_3 = G \mid G)P(G \mid J_1 = G, J_2 = G) \\ &\quad + P(J_3 = G \mid I)P(I \mid J_1 = G, J_2 = G) \\ &= (0.7)\frac{343}{355} + (0.2)\frac{12}{355} \\ &= \frac{242.5}{355} \\ &\approx 0.6831. \end{aligned}$$

Hence,

$$\boxed{P(J_3 = G \mid J_1 = G, J_2 = G) \approx 0.6831}.$$

(b) judges 1 and 2 cast one guilty and one not guilty vote, We want to find

$$P(J_3 = G \mid J_1 = G, J_2 = I).$$

Given that the defendant is guilty,

$$P(J_1 = G, J_2 = I \mid G) = (0.7)(0.3) = 0.21.$$

Given that the defendant is innocent,

$$P(J_1 = G, J_2 = I \mid I) = (0.2)(0.8) = 0.16.$$

By Bayes' theorem,

$$P(G \mid J_1 = G, J_2 = I) = \frac{(0.21)(0.7)}{(0.21)(0.7) + (0.16)(0.3)} = \frac{0.147}{0.195} = \frac{49}{65}.$$

Thus,

$$P(I \mid J_1 = G, J_2 = I) = 1 - \frac{49}{65} = \frac{16}{65}.$$

Therefore,

$$\begin{aligned} P(J_3 = G \mid J_1 = G, J_2 = I) &= P(J_3 = G \mid G)P(G \mid J_1 = G, J_2 = I) \\ &\quad + P(J_3 = G \mid I)P(I \mid J_1 = G, J_2 = I) \\ &= (0.7)\frac{49}{65} + (0.2)\frac{16}{65} \\ &= \boxed{\frac{37.5}{65} \approx 0.5769}. \end{aligned}$$

(c) judges 1 and 2 both cast not guilty votes. We want to find

$$P(J_3 = G \mid J_1 = I, J_2 = I).$$

Given that the defendant is guilty,

$$P(J_1 = I, J_2 = I \mid G) = (0.3)^2 = 0.09.$$

Given that the defendant is innocent,

$$P(J_1 = I, J_2 = I \mid I) = (0.8)^2 = 0.64.$$

By Bayes' theorem,

$$P(G \mid J_1 = I, J_2 = I) = \frac{(0.09)(0.7)}{(0.09)(0.7) + (0.64)(0.3)} = \frac{0.063}{0.255} = \frac{21}{85}.$$

Thus,

$$P(I \mid J_1 = I, J_2 = I) = 1 - \frac{21}{85} = \frac{64}{85}.$$

Therefore,

$$\begin{aligned} P(J_3 = G \mid J_1 = I, J_2 = I) &= P(J_3 = G \mid G)P(G \mid J_1 = I, J_2 = I) \\ &\quad + P(J_3 = G \mid I)P(I \mid J_1 = I, J_2 = I) \\ &= (0.7)\frac{21}{85} + (0.2)\frac{64}{85} \\ &= \boxed{\frac{27.5}{85} \approx 0.3235}. \end{aligned}$$

Observe that as the evidence shifts based on other judges, Judge 3's confidence changes.

2. Let X and Y be random variables with finite variances

$$\mathrm{Var}(X) = \sigma_X^2 \qquad \text{and} \qquad \mathrm{Var}(Y) = \sigma_Y^2.$$

(a) As something unrelated to X and Y – but will be used later on in (d) – prove that if $\mathrm{Var}(Z) = 0$, then

$$P\{Z = \mathbb{E}[Z]\} = 1.$$

By definition of variance,

$$\mathrm{Var}(Z) = \mathbb{E}\left[(Z - \mathbb{E}[Z])^2\right].$$

Since $\mathrm{Var}(Z) = 0$, we have

$$\mathbb{E}\left[(Z - \mathbb{E}[Z])^2\right] = 0.$$

Now, $(Z - \mathbb{E}[Z])^2 \geq 0$. A nonnegative random variable can have expectation 0 only if it is equal to 0 almost surely.

Aside. The phrase "almost surely": Let $W = (Z - \mathbb{E}[Z])^2$ be non-negative. Piazza post 53 https://piazza.com/class/ms1ll7021ih5yc/post/53_f3 is pointing to the fact that if the expectation – written as an integral – is zero does not mean that the integrand is zero.
But let's say the integrand is non-zero. If W happens to be non-zero with probability > 0 on some weird situation, then the expectation will be non-zero. So the probability that this happens is 0, and thus W = 0, almost surely.

Hence,

$$(Z - \mathbb{E}[Z])^2 = 0$$

Therefore,

$$Z = \mathbb{E}[Z] \text{(almost surely)}$$

which means

$$\boxed{P\{Z = \mathbb{E}[Z]\} = 1}.$$

(b) Using the fact that variance is always nonnegative, show that

$$0 \leq \operatorname{Var}\left(\frac{X}{\sigma_X} + \frac{Y}{\sigma_Y}\right)$$

implies that

$$-1 \leq \operatorname{Corr}(X, Y).$$

Using the properties of variance,

$$\begin{aligned} 0 &\leq \operatorname{Var}\left(\frac{X}{\sigma_X} + \frac{Y}{\sigma_Y}\right) \\ &= \frac{\operatorname{Var}(X)}{\sigma_X^2} + \frac{\operatorname{Var}(Y)}{\sigma_Y^2} + \frac{2\operatorname{Cov}(X, Y)}{\sigma_X \sigma_Y}. \end{aligned}$$

Since $\sigma_X^2 = \operatorname{Var}(X)$ and $\sigma_Y^2 = \operatorname{Var}(Y)$,

$$0 \leq 1 + 1 + 2\frac{\operatorname{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

Using

$$\operatorname{Corr}(X, Y) = \frac{\operatorname{Cov}(X, Y)}{\sigma_X \sigma_Y},$$

we obtain

$$0 \leq 2 + 2\operatorname{Corr}(X, Y).$$

and hence

$$\boxed{-1 \leq \operatorname{Corr}(X, Y)}.$$

(c) Similarly, use

$$0 \leq \operatorname{Var}\left(\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y}\right)$$

to prove that

$$-1 \leq \operatorname{Corr}(X, Y) \leq 1.$$

Using the properties of variance,

$$\begin{aligned} 0 &\leq \operatorname{Var}\left(\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y}\right) \\ &= \frac{\operatorname{Var}(X)}{\sigma_X^2} + \frac{\operatorname{Var}(Y)}{\sigma_Y^2} - \frac{2\operatorname{Cov}(X, Y)}{\sigma_X \sigma_Y}. \end{aligned}$$

Since $\sigma_X^2 = \operatorname{Var}(X)$ and $\sigma_Y^2 = \operatorname{Var}(Y)$,

$$0 \leq 1 + 1 - 2\frac{\operatorname{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

Using

$$\operatorname{Corr}(X, Y) = \frac{\operatorname{Cov}(X, Y)}{\sigma_X \sigma_Y},$$

we obtain

$$0 \leq 2 - 2\,\mathrm{Corr}(X, Y).$$

and hence

$$\boxed{\mathrm{Corr}(X, Y) \leq 1}.$$

(d) Using the result from part (a), prove that if

$$\mathrm{Corr}(X, Y) = 1$$

or

$$\mathrm{Corr}(X, Y) = -1,$$

then X and Y must satisfy a linear relationship of the form

$$Y = a + bX,$$

where $b > 0$ when $\mathrm{Corr}(X, Y) = 1$ and $b < 0$ when $\mathrm{Corr}(X, Y) = -1$.
If $\mathrm{Corr}(X, Y) = 1$, then equality holds in the variance inequality proved earlier, and hence

$$\mathrm{Var}\left(\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y}\right) = 0.$$

By part (a), if a random variable has variance 0, then it is equal to its expectation almost surely. Hence,

$$\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y} = \mathbb{E}\left[\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y}\right]$$

Therefore,

$$Y = \frac{\sigma_Y}{\sigma_X} X + \sigma_Y \left(\frac{\mathbb{E}[Y]}{\sigma_Y} - \frac{\mathbb{E}[X]}{\sigma_X}\right).$$

Thus,

$$Y = a + bX, \qquad b = \frac{\sigma_Y}{\sigma_X} > 0$$

since Corr(X,y) is well defined. Similarly, if $\mathrm{Corr}(X, Y) = -1$, then

$$\mathrm{Var}\left(\frac{X}{\sigma_X} + \frac{Y}{\sigma_Y}\right) = 0.$$

Hence,

$$\frac{X}{\sigma_X} + \frac{Y}{\sigma_Y} = \mathbb{E}\left[\frac{X}{\sigma_X} + \frac{Y}{\sigma_Y}\right]$$

$$Y = -\frac{\sigma_Y}{\sigma_X} X + \sigma_Y \left(\frac{\mathbb{E}[X]}{\sigma_X} + \frac{\mathbb{E}[Y]}{\sigma_Y}\right).$$

Thus,

$$Y = a + bX, \qquad b = -\frac{\sigma_Y}{\sigma_X} < 0.$$

Therefore, in either case, X and Y satisfy a linear relationship

$$\boxed{Y = a + bX},$$

with $b > 0$ when $\mathrm{Corr}(X, Y) = 1$ and $b < 0$ when $\mathrm{Corr}(X, Y) = -1$.

3. Consider n independent trials, each of which results in one of the three outcomes 1, 2, or 3, with corresponding probabilities p_1, p_2, and p_3, where

$$p_1 + p_2 + p_3 = 1.$$

Let N_i denote the number of trials that result in outcome i.

Find

$$\mathbb{E}[N_i]$$

For each $j = 1, 2, \ldots, n$, define the indicator random variable

$$I_{ij} = \begin{cases} 1, & \text{if the } j\text{-th trial results in outcome } i, \\ 0, & \text{otherwise.} \end{cases}$$

Since N_i is the total number of trials that result in outcome i,

$$N_i = \sum_{j=1}^{n} I_{ij}.$$

For each trial j,

$$\mathbb{P}(I_{ij} = 1) = p_i, \qquad \mathbb{P}(I_{ij} = 0) = 1 - p_i.$$

Therefore,

$$\begin{aligned} \mathbb{E}[I_{ij}] &= 1 \cdot \mathbb{P}(I_{ij} = 1) + 0 \cdot \mathbb{P}(I_{ij} = 0) \\ &= 1 \cdot p_i + 0 \cdot (1 - p_i) \\ &= p_i. \end{aligned}$$

Using linearity of expectation,

$$\begin{aligned} \mathbb{E}[N_i] &= \mathbb{E}\left[\sum_{j=1}^{n} I_{ij}\right] \\ &= \sum_{j=1}^{n} \mathbb{E}[I_{ij}] \\ &= \sum_{j=1}^{n} p_i \\ &= \boxed{np_i}. \end{aligned}$$

4. Let (X, Y, Z) have joint probability density

$$f_{X,Y,Z}(x, y, z) = \begin{cases} c(x + z), & 0 < x < y < z < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Determine the constant c.

Since $f_{X,Y,Z}$ is a probability density function, its integral over the entire support must be equal to 1. Hence,

$$\int_0^1 \int_0^z \int_{y=x}^{z} c(x + z)\, dy\, dx\, dz = 1.$$

Integrating first with respect to y,

$$c \int_0^1 \int_0^z (x+z)(z-x)\, dx\, dz = 1.$$

Since

$$(x+z)(z-x) = z^2 - x^2,$$

we get

$$c \int_0^1 \int_0^z (z^2 - x^2)\, dx\, dz = c \int_0^1 \frac{2z^3}{3}\, dz = \frac{c}{6}.$$

Therefore,

$$\boxed{c = 6}.$$

(b) Compute

$$P(X + Y < Z).$$

Hint: Refer to example 4.3c from the textbook

We need the region satisfying both

$$0 < x < y < z < 1$$

and

$$x + y < z.$$

For fixed z, since $y > x$ and $y < z - x$, we must have

$$x < z - x,$$

or equivalently,

$$0 < x < \frac{z}{2}.$$

Thus,

$$P(X + Y < Z) = 6 \int_0^1 \int_0^{z/2} \int_x^{z-x} (x+z)\, dy\, dx\, dz.$$

Integrating with respect to y,

$$= 6 \int_0^1 \int_0^{z/2} (x+z)(z-2x)\, dx\, dz.$$

Evaluating the remaining integrals gives

$$P(X + Y < Z) = 6 \int_0^1 \frac{7z^3}{24}\, dz = \frac{7}{16}.$$

Hence,

$$\boxed{P(X + Y < Z) = \frac{7}{16}}.$$

(c) Compute

$$E[XZ].$$

Using the definition of expectation for jointly continuous random variables,

$$E[XZ] = \iiint xz\, f_{X,Y,Z}(x,y,z)\, dydxdz.$$

Therefore,

$$E[XZ] = 6\int_0^1 \int_0^z \int_x^z xz(x+z)\, dy\, dx\, dz.$$

Integrating first with respect to y,

$$E[XZ] = 6\int_0^1 \int_0^z xz(x+z)(z-x)\, dx\, dz.$$

Since

$$(x+z)(z-x) = z^2 - x^2,$$

this becomes

$$6\int_0^1 \int_0^z xz(z^2-x^2)\, dx\, dz.$$

Evaluating,

$$E[XZ] = 6\int_0^1 \frac{z^5}{4}\, dz = \frac{1}{4}.$$

Hence,

$$\boxed{E[XZ] = \frac{1}{4}}.$$

5. Let $X \geq 0$ be a random variable satisfying

$$E[X] = 4, \qquad E[X^2] = 24, \qquad E[X^3] = 192.$$

Use Markov's inequality, applied appropriately to X, X^2, and X^3, to obtain a tight bound from the given information on

$$P(X \geq 10).$$

Since $X \geq 0$, Markov's inequality gives

$$P(X \geq a) \leq \frac{E[X]}{a}.$$

Applying it directly to X,

$$P(X \geq 10) \leq \frac{E[X]}{10} = \frac{4}{10} = 0.4.$$

We can also apply Markov's inequality to the nonnegative random variable X^2. Since

$$X \geq 10 \quad \Longleftrightarrow \quad X^2 \geq 100,$$

we get

$$P(X \geq 10) = P(X^2 \geq 100) \leq \frac{E[X^2]}{100} = \frac{24}{100} = 0.24.$$

Similarly, applying Markov's inequality to X^3,

$$P(X \geq 10) = P(X^3 \geq 1000) \leq \frac{E[X^3]}{1000} = \frac{192}{1000} = 0.192.$$

Therefore, the tightest of the three bounds is

$$\boxed{P(X \geq 10) \leq 0.192}.$$

A linear combination, or more generally, a non-linear combination, of the given expectations is possible to get an even tighter bound. This is a much more open ended problem as also noted in the Piazza post.

6. Let $X > 0$ be a random variable satisfying

$$E[X] = \mu.$$

For some constant $a > 0$, define

$$g(x) = x + \frac{a}{x}.$$

(a) Use Jensen's inequality to obtain a bound, in terms of μ and a, for

$$E\left[X + \frac{a}{X}\right].$$

For $x > 0$,

$$g(x) = x + \frac{a}{x}.$$

Since

$$g''(x) = \frac{2a}{x^3} > 0,$$

the function g is convex on $(0, \infty)$.
Therefore, by Jensen's inequality,

$$E[g(X)] \geq g(E[X]).$$

Using $E[X] = \mu$,

$$E\left[X + \frac{a}{X}\right] \geq \mu + \frac{a}{\mu}.$$

Hence,

$$\boxed{E\left[X + \frac{a}{X}\right] \geq \mu + \frac{a}{\mu}}.$$

(b) If μ may also be chosen freely, determine the value of μ to minimize the Jensen bound.

From part (a), the Jensen bound is

$$\mu + \frac{a}{\mu}, \qquad \mu > 0.$$

To minimize this expression, differentiate with respect to μ:

$$\frac{d}{d\mu}\left(\mu + \frac{a}{\mu}\right) = 1 - \frac{a}{\mu^2}.$$

Setting the derivative equal to zero gives

$$1 - \frac{a}{\mu^2} = 0,$$

so

$$\mu^2 = a.$$

Since $\mu > 0$,

$$\boxed{\mu = \sqrt{a}}.$$

At this value, the minimum Jensen bound is

$$\sqrt{a} + \frac{a}{\sqrt{a}} = 2\sqrt{a}.$$

Thus,

$$\boxed{\min_{\mu>0} \left(\mu + \frac{a}{\mu}\right) = 2\sqrt{a}}.$$

7. Let $X_1, \ldots, X_{25}$ be identically distributed random variables with

$$E[X_i] = 100, \qquad \mathrm{Var}(X_i) = 400.$$

Suppose that, for $i \neq j$,

$$\mathrm{Cov}(X_i, X_j) = 40.$$

Let

$$\overline{X} = \frac{1}{25} \sum_{i=1}^{25} X_i.$$

(a) Use Chebyshev's inequality to obtain a lower bound for

$$P(92 < \overline{X} < 108).$$

First,

$$E[\overline{X}] = \frac{1}{25} \sum_{i=1}^{25} E[X_i] = 100.$$

To compute the variance, we use

$$\mathrm{Var}\left(\sum_{i=1}^{25} X_i\right) = \sum_{i=1}^{25} \mathrm{Var}(X_i) + \sum_{i \neq j} \mathrm{Cov}(X_i, X_j).$$

Hence,

$$\begin{aligned} \mathrm{Var}(\overline{X}) &= \frac{1}{25^2} \left(25(400) + 25(24)(40)\right) \\ &= \frac{34000}{625} = 54.4. \end{aligned}$$

Now,

$$P(92 < \overline{X} < 108) = P(|\overline{X} - 100| < 8).$$

By Chebyshev's inequality,

$$P(|\overline{X} - 100| \geq 8) \leq \frac{\text{Var}(\overline{X})}{8^2} = \frac{54.4}{64} = 0.85.$$

Therefore,

$$\boxed{P(92 < \overline{X} < 108) \geq 0.15}.$$

(b) Compare this bound with the one obtained if the X_i's were independent. Expand on your observations on the the two differing bounds.

If the X_i's were independent, then all the covariance terms would be zero. Thus,

$$\text{Var}(\overline{X}) = \frac{1}{25^2}(25(400)) = 16.$$

Applying Chebyshev's inequality again,

$$P(|\overline{X} - 100| \geq 8) \leq \frac{16}{64} = \frac{1}{4}.$$

Hence,

$$\boxed{P(92 < \overline{X} < 108) \geq \frac{3}{4}}.$$

Thus, the Chebyshev lower bound increases from

$$0.15 \qquad \text{to} \qquad 0.75$$

when the variables are independent.

The reason the lower bound tells us more about the probabilistic event is obviously because the variance has dropped dramatically from $\approx$54 to 16. When the observations are independent, averaging reduces variability. If this was a real experiment with 25 different students, we would not want the students to "copy" the results from one another (larger covariance), but be more "independent" (covariance = 0).

CS 215: Task 04: Due: 30-Aug; time as announced on Piazza. No extensions.

- Your answers should be in your handwriting. Try to be neat and clear. If an LLM can read your handwriting, we might be able to.
- Please write the honor code statement only if it is true. If you used any source (person or thing) explicitly state it. This is an individual assignment.

Teaching-team hint.

- When we say, 'derive' or 'show', simply put 2-3 important steps in the derivation in the space provided. This is practice for your midsemester exam, where we may not be able to give extra space. You have to decide what is important. For the homework, please attach extra paper for your records and for you to scan in the upload.

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1. $U \sim \text{Uniform}(0,1)$ and $a < b$, define
$$U' = a + (b-a)U.$$
(a) Show that U' is uniformly distributed on (a,b) by working with the cumulative distribution function of U'.

Teaching-team hint. Do not use any shortcut such as the Leibniz rule discussed in class.

(b) Derive the moment-generating function of U'.

(c) Use the MGF to compute $E(U')$ and $\mathrm{Var}(U')$.

2. $X \sim \text{Hypergeometric}(M, N, K)$, thus,

$$P(X = x \mid M, N, K) = \frac{\binom{M}{x}\binom{N-M}{K-x}}{\binom{N}{K}}, \qquad x = 0, 1, \ldots, K.$$

In class we discussed the relationship of the expectation and the variance of X with the binomial distribution. Now, show, using Stirling's approximation, viz., $n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$, that if $N \to \infty$, $M \to \infty$, and

$$\frac{M}{N} \to p, \qquad 0 < p < 1,$$

then, for fixed K and x,

$$P(X = x \mid N, M, K) \longrightarrow \binom{K}{x} p^x (1-p)^{K-x}.$$

3. Let X be a random variable with moment-generating function

$$M_X(t) = E(e^{tX}), \qquad -h < t < h.$$

Prove that

$$P(X \geq a) \leq e^{-at} M_X(t), \qquad 0 < t < h.$$

4. $X \sim \text{Binomial}(n, p)$. Our goal is to determine the value of k for which $P(X = k)$, is maximized.
   (a) Show that
   $$P(X = k+1) = \frac{p}{1-p}\frac{n-k}{k+1}P(X = k), \qquad k = 0, 1, \ldots, n-1.$$

   (b) Show that as k goes from 0 to n, the sequence $P(X = k)$ first increases and then decreases. Determine the value of k for which $P(X = k)$ is maximum. Discuss when two values tie.

5. A stick of length 1 metre is broken at two points chosen independently and uniformly at random along its length. Determine the expected lengths of the largest and smallest pieces.

   Teaching-team hint. Let the two breaking points, in increasing order, be $U_{(1)}$ and $U_{(2)}$. Express the three piece lengths in terms of these variables. Reason on the constraints in two dimensions.

6. Let $U \sim \text{Uniform}(0, 1)$ and define

$$X = U(1 - U).$$

Find the cumulative distribution function and probability density function of X.

7. Let $X \sim \text{Binomial}(n, p)$. Define

$$Y = \frac{1}{X + 1}.$$

(a) Show that

$$E\left(\frac{1}{X + 1}\right) = \frac{1 - (1 - p)^{n+1}}{(n + 1)p}.$$

(b) Deduce a closed-form expression for

$$E\left(\frac{X}{X+1}\right).$$

CS 215: Task 02: Due: 15-Aug; time as announced on Piazza. No extensions.

- Your answers should be in your handwriting. Try to be neat and clear. If an LLM can read your handwriting, we might be able to.
- Please write the honor code statement only if it is true. If you used any source (person or thing) explicitly state it. This is an individual assignment.

Teaching-team notes.

- When we say, 'derive' or 'show', simply put 2-3 important steps in the derivation in the space provided. This is practice for your midsemester exam, where we may not be able to give extra space. You have to decide what is important. For the homework, please attach extra paper for your records and for you to scan in the upload.

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0. Please write the honor code in your handwriting, with the understanding that it is true.

Solution:

1. $U \sim \text{Uniform}(0,1)$ and $a < b$, define
$$U' = a + (b-a)U.$$
(a) Show that U' is uniformly distributed on (a,b) by working with the cumulative distribution function of U'.

Teaching-team notes. Do not use any shortcut such as the Leibniz rule discussed in class.

Solution: For $a < x < b$,
$$\begin{aligned} P(U' \le x) &= P\left(a + (b-a)U \le x\right) \\ &= P\left(U \le \frac{x-a}{b-a}\right) \\ &= \frac{x-a}{b-a}. \end{aligned}$$
Therefore, the cumulative distribution function of U' is
$$F_{U'}(x) = \begin{cases} 0, & x \le a, \\ \dfrac{x-a}{b-a}, & a < x < b, \\ 1, & x \ge b. \end{cases}$$
This is precisely the CDF of a Uniform(a,b) random variable. Hence,
$$\boxed{U' \sim \text{Uniform}(a,b).}$$

(b) Derive the moment-generating function of U'.

Solution: Since $U' \sim \text{Uniform}(a, b)$, its density is

$$f_{U'}(x) = \frac{1}{b-a}, \qquad a < x < b.$$

Hence, for $t \neq 0$,

$$\begin{aligned} M_{U'}(t) &= E(e^{tU'}) \\ &= \frac{1}{b-a} \int_a^b e^{tx}\, dx \\ &= \frac{1}{b-a} \left[\frac{e^{tx}}{t} \right]_a^b \\ &= \frac{e^{bt} - e^{at}}{(b-a)t}. \end{aligned}$$

At $t = 0$, by continuity,

$$M_{U'}(0) = 1.$$

Therefore,

$$\boxed{M_{U'}(t) = \begin{cases} \dfrac{e^{bt} - e^{at}}{(b-a)t}, & t \neq 0, \\ 1, & t = 0. \end{cases}}$$

(c) Use the MGF to compute $E(U')$ and $\text{Var}(U')$.

Solution: Using the Taylor expansion of the MGF around $t = 0$,

$$M_{U'}(t) = 1 + E(U')t + \frac{E((U')^2)}{2!}t^2 + \cdots.$$

Expanding the MGF,

$$M_{U'}(t) = \frac{e^{bt} - e^{at}}{(b-a)t} = 1 + \frac{a+b}{2}t + \frac{a^2 + ab + b^2}{6}t^2 + \cdots.$$

Therefore,

$$E(U') = M'_{U'}(0) = \frac{a+b}{2},$$

and

$$E((U')^2) = M''_{U'}(0) = \frac{a^2 + ab + b^2}{3}.$$

Hence,

$$\begin{aligned} \text{Var}(U') &= E((U')^2) - [E(U')]^2 \\ &= \frac{a^2 + ab + b^2}{3} - \left(\frac{a+b}{2} \right)^2 \\ &= \frac{(b-a)^2}{12}. \end{aligned}$$

Thus,

$$\boxed{E(U') = \frac{a+b}{2}}, \qquad \boxed{\text{Var}(U') = \frac{(b-a)^2}{12}}.$$

2. $X \sim \text{Hypergeometric}(M, N, K)$, thus,

$$P(X = x \mid M, N, K) = \frac{\binom{M}{x}\binom{N-M}{K-x}}{\binom{N}{K}}, \qquad x = 0, 1, \ldots, K.$$

In class we discussed the relationship of the expectation and the variance of X with the binomial distribution. Now, show, using Stirling's approximation, viz., $n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$, that if $N \to \infty$, $M \to \infty$, and

$$\frac{M}{N} \to p, \qquad 0 < p < 1,$$

then, for fixed K and x,

$$P(X = x \mid M, N, K) \longrightarrow \binom{K}{x} p^x (1-p)^{K-x}.$$

Solution: For a fixed integer r and $n \to \infty$, Stirling's approximation gives

$$n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n,$$

and consequently

$$\binom{n}{r} \sim \frac{n^r}{r!}.$$

Since

$$\frac{M}{N} \to p,$$

we have

$$M \sim pN$$

and

$$N - M \sim (1-p)N.$$

For fixed x and K,

$$\binom{M}{x} \sim \frac{M^x}{x!},$$

$$\binom{N-M}{K-x} \sim \frac{(N-M)^{K-x}}{(K-x)!},$$

and

$$\binom{N}{K} \sim \frac{N^K}{K!}.$$

Therefore,

$$\begin{aligned} P(X = x \mid M, N, K) &= \frac{\binom{M}{x}\binom{N-M}{K-x}}{\binom{N}{K}} \\ &\sim \frac{\dfrac{M^x}{x!}\dfrac{(N-M)^{K-x}}{(K-x)!}}{\dfrac{N^K}{K!}}. \end{aligned}$$

Hence,

$$P(X = x \mid M, N, K) \sim \frac{K!}{x!(K-x)!}\left(\frac{M}{N}\right)^x \left(\frac{N-M}{N}\right)^{K-x}.$$

$$\frac{N-M}{N} = 1 - \frac{M}{N} \to 1 - p.$$

Thus,

$$\boxed{P(X = x \mid M, N, K) \longrightarrow \binom{K}{x} p^x (1-p)^{K-x}.}$$

Therefore, for fixed sample size K, the hypergeometric distribution approaches the Binomial(K, p) distribution as the population size becomes large with the population proportion of successes approaching p.

3. Let X be a random variable with moment-generating function

$$M_X(t) = E(e^{tX}), \qquad -h < t < h.$$

Prove that

$$P(X \geq a) \leq e^{-at} M_X(t), \qquad 0 < t < h.$$

Solution: For $t > 0$, since the exponential function is increasing, the event

$$\{X \geq a\}$$

is the same as the event

$$\{e^{tX} \geq e^{ta}\}.$$

Applying Markov's inequality to the nonnegative random variable e^{tX},

$$\begin{aligned} P(X \geq a) &= P(e^{tX} \geq e^{ta}) \\ &\leq \frac{E(e^{tX})}{e^{ta}} \\ &= e^{-at} M_X(t). \end{aligned}$$

Teaching-team notes. An alternate proof appears below.
For $t > 0$, on the event $\{X \geq a\}$,

$$tX \geq ta.$$

Since the exponential function is increasing,

$$e^{tX} \geq e^{ta}.$$

Let

$$I_{\{X \geq a\}} = \begin{cases} 1, & X \geq a, \\ 0, & X < a. \end{cases}$$

If $X \geq a$, then

$$1 \leq e^{-at} e^{tX}$$

while if $X < a$, then

$$0 \leq e^{-at} e^{tX}$$

Therefore,

$$I_{\{X \geq a\}} \leq e^{-at} e^{tX}.$$

Taking expectations on both sides,

$$\begin{aligned} P(X \geq a) &= E\left[I_{\{X \geq a\}}\right] \\ &\leq e^{-at} E(e^{tX}) \\ &= e^{-at} M_X(t). \end{aligned}$$

Hence,

4. $X \sim \text{Binomial}(n, p)$. Our goal is to determine the value of k for which $P(X = k)$, is maximized.

   (a) Show that

$$P(X = k+1) = \frac{p}{1-p}\frac{n-k}{k+1}P(X = k), \qquad k = 0, 1, \ldots, n-1.$$

   Solution: For a binomial random variable,

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

   Therefore,

$$P(X = k+1) = \binom{n}{k+1} p^{k+1} (1-p)^{n-k-1}.$$

   Taking the ratio,

$$\begin{aligned} \frac{P(X = k+1)}{P(X = k)} &= \frac{\binom{n}{k+1}}{\binom{n}{k}} \frac{p}{1-p} \\ &= \frac{n-k}{k+1}\frac{p}{1-p}. \end{aligned}$$

$$\boxed{P(X = k+1) = \frac{p}{1-p}\frac{n-k}{k+1}P(X = k).}$$

   (b) Assume $0 < p < 1$, and n is sufficiently large so that

$$\frac{1}{n+1} < p < \frac{n}{n+1}.$$

   Show that, as k goes from 0 to n, the sequence $P(X = k)$ first increases and then decreases, except possibly for a plateau at the maximum. Determine the value of k for which $P(X = k)$ is maximum. Discuss when two values tie.

   ~~Show that as k goes from 0 to n, the sequence $P(X = k)$ first increases and then decreases. Determine the value of k for which $P(X = k)$ is maximum. Discuss when two values tie.~~

   Solution: From part (a),

$$\frac{P(X = k+1)}{P(X = k)} = \frac{p}{1-p}\frac{n-k}{k+1}.$$

   The sequence is increasing when

$$\frac{P(X = k+1)}{P(X = k)} > 1.$$

   Since $0 < p < 1$, this is equivalent to

$$p(n-k) > (1-p)(k+1).$$

   Simplifying,

$$np - pk > k + 1 - pk - p,$$

   so

$$(n+1)p > k + 1.$$

   Thus,

$$P(X = k+1) > P(X = k) \quad \Longleftrightarrow \quad k + 1 < (n+1)p.$$

Similarly,

$$P(X = k+1) < P(X = k) \quad \Longleftrightarrow \quad k+1 > (n+1)p.$$

Therefore, the probabilities first increase and then decrease, except possibly for the plateau.
Let

$$m = \lfloor (n+1)p \rfloor .$$

If $(n+1)p$ is not an integer, then

$$\boxed{m = \lfloor (n+1)p \rfloor}$$

is the unique value of k at which the maximum occurs.
If $(n+1)p$ is an integer, say $(n+1)p = m$, then

$$P(X = m) = P(X = m-1),$$

so there are two modes, $m-1$ and m.
In either case, the largest integer less than or equal to $(n+1)p$ is a value of k at which the maximum is attained.
Hence,

$$\boxed{\begin{cases} \lfloor (n+1)p \rfloor , & \text{if } (n+1)p \notin \mathbb{Z}, \\ (n+1)p - 1 \text{ and } (n+1)p, & \text{if } (n+1)p \in \mathbb{Z}. \end{cases}}$$

5. A stick of length 1 metre is broken at two points chosen independently and uniformly at random along its length. Determine the expected lengths of the largest and smallest pieces.

Teaching-team notes. Let the two breaking points, in increasing order, be $U_{(1)}$ and $U_{(2)}$. Express the three piece lengths in terms of these variables. Reason on the constraints in two dimensions.

Solution: Let the two breaking points, in increasing order, be $U_{(1)}$ and $U_{(2)}$. The three piece lengths are

$$L_1 = U_{(1)}, \qquad L_2 = U_{(2)} - U_{(1)}, \qquad L_3 = 1 - U_{(2)}.$$

Consider the unit square corresponding to the two ordered breaking points $(U_{(1)}, U_{(2)})$. Since $U_{(1)} < U_{(2)}$, the possible values lie in the triangle above the line

$$U_{(2)} = U_{(1)},$$

which has area $1/2$. The two problems of expected lengths will require how this feasible area is subdivided.

Smallest piece

Let S denote the smallest of the three pieces. We cannot have all 3 pieces greater than $1/3$, so we consider $0 < x < 1/3$, and look for

$$P(S > x) = P(L_1 > x, L_2 > x, L_3 > x).$$

We sketch three relevant boundaries to find the allowed region. Pick an $x < 1/3$.

- Now $S > x$ means that the first boundary has $U_{(1)} > x$. Think of this as a vertical line.

- Another boundary comes from the horizontal line $U_{(2)} < 1 - x$ since $L_3 = 1 - U_{(2)}$ and $L_3 > x$.
- The third boundary is the inclined line $U_{(2)} = U_{(1)} + x$ with "y-intercept" as the value x. (remember, the axes are represented by $(U_{(1)}, U_{(2)})$).

These three boundaries carve out a small triangle inside the allowed triangle of area $1/2$. Checking out the coordinates of this triangle we get a right triangle with two perpendicular sides with length $(1 - x) - 2x = 1 - 3x$. Hence this area $(1 - 3x)^2/2$, while the entire ordered region has area $1/2$. Hence,

$$P(S > x) = (1 - 3x)^2, \qquad 0 < x < \frac{1}{3}.$$

The cumulative distribution function is therefore

$$F_S(x) = P(S \leq x) = 1 - P(S > x) = 1 - (1 - 3x)^2.$$

Thus, for $0 < x < 1/3$,

$$F_S(x) = 6x - 9x^2.$$

Differentiating,

$$f_S(x) = 6 - 18x, \qquad 0 < x < \frac{1}{3}.$$

Therefore,

$$\begin{aligned} E(S) &= \int_0^{1/3} x f_S(x)\, dx \\ &= \int_0^{1/3} x(6 - 18x)\, dx \\ &= \int_0^{1/3} (6x - 18x^2)\, dx \\ &= \left[3x^2 - 6x^3\right]_0^{1/3} \\ &= \frac{1}{9}. \end{aligned}$$

Largest piece

Now let M denote the largest piece. We will look for $P(M \leq x)$.

1. For $0 < x < 1$,

$$P(M \leq x) = 0, \qquad x < \frac{1}{3}.$$

2. For $1/3 \leq x \leq 1/2$, the three boundary conditions result in
   - $U_{(1)} \leq x$
   - $U_{(2)} \geq 1 - x$
   - $U_{(2)} \leq U_{(1)} + x$

   This results in a triangle with area $0.5(3x - 1)^2$. Since the entire ordered region has area $1/2$,

$$F_M(x) = P(M \leq x) = \frac{\frac{1}{2}(3x - 1)^2}{\frac{1}{2}} = (3x - 1)^2, \qquad \frac{1}{3} \leq x \leq \frac{1}{2}.$$

3. For $\frac{1}{2} \le x < 1$, two different pieces cannot both exceed x. Indeed, if for example

$$L_1 > x \qquad \text{and} \qquad L_2 > x,$$

then

$$L_1 + L_2 > 2x \ge 1,$$

which is impossible since

$$L_1 + L_2 + L_3 = 1$$

and $L_3 \ge 0$.

Thus, for $x \ge 1/2$, the three events

$$\{L_1 > x\}, \qquad \{L_2 > x\}, \qquad \{L_3 > x\}$$

are mutually disjoint.

Since

$$M = \max(L_1, L_2, L_3),$$

we have

$$\{M > x\} = \{L_1 > x\} \cup \{L_2 > x\} \cup \{L_3 > x\}.$$

Therefore,

$$\begin{aligned} P(M > x) &= P(L_1 > x) + P(L_2 > x) + P(L_3 > x) \\ &= 3(1-x)^2. \end{aligned}$$

In the last step, we have determined the individual densities again by considering the unit square and figuring out the boundaries as in the previous cases. Specifically $P(L_1 \ge x)$ will simply create a corner triangle of side lengths $(1-x)$, and thus the probability is $(1-x)^2$. By symmetry, $P(L_3 \ge x)$ is the same. For the middle piece, $P(L_2 > w) = P((U_{(2)} - U_{(1)}) > w)$, and again using the geometry of the unit square, we have

$$P(L_2 > w) = (1-w)^2, \qquad 0 < w < 1.$$

Returning to the largest piece M, we thus have

$$F_M(x) = P(M \le x) = 1 - 3(1-x)^2, \qquad \frac{1}{2} \le x < 1.$$

Differentiating the cumulative distribution functions for the three parts, we obtain

$$f_M(x) = \begin{cases} 18x - 6, & \frac{1}{3} < x < \frac{1}{2}, \\ 6(1-x), & \frac{1}{2} < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Therefore,

$$\begin{aligned} E(M) &= \int_{1/3}^{1/2} x(18x-6)\,dx + \int_{1/2}^{1} x\,6(1-x)\,dx \\ &= \left[6x^3 - 3x^2\right]_{1/3}^{1/2} + \left[3x^2 - 2x^3\right]_{1/2}^{1} \\ &= \frac{1}{9} + \frac{1}{2} \\ &= \frac{11}{18}. \end{aligned}$$

Thus,

$$\boxed{E(M) = \frac{11}{18}}.$$

6. Let $U \sim \text{Uniform}(0,1)$ and define

$$X = U(1-U).$$

Find the cumulative distribution function and probability density function of X.

Solution: Since

$$X = U(1-U) = \frac{1}{4} - \left(U - \frac{1}{2}\right)^2,$$

the range of X is

$$0 \le x \le \frac{1}{4}.$$

For $0 < x < 1/4$,

$$\begin{aligned} F_X(x) &= P(U(1-U) \le x) \\ &= P\left(U^2 - U + x \ge 0\right). \end{aligned}$$

The roots of

$$u^2 - u + x = 0$$

are

$$u = \frac{1 \pm \sqrt{1-4x}}{2}.$$

We will use the roots to find the region where the quadratic is non-negative and also apply the constraint on $U \in [0,1]$.

Therefore,

$$U(1-U) \le x$$

when

$$0 \le U \le \frac{1-\sqrt{1-4x}}{2}$$

or

$$\frac{1+\sqrt{1-4x}}{2} \le U \le 1.$$

Since U is uniform on $(0,1)$, we simply have to add up the lengths:

$$\begin{aligned} F_X(x) &= \frac{1-\sqrt{1-4x}}{2} + (1 - \frac{1+\sqrt{1-4x}}{2}) \\ &= 1 - \sqrt{1-4x}. \end{aligned}$$

Hence,

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 1 - \sqrt{1-4x}, & 0 \le x \le \frac{1}{4}, \\ 1, & x > \frac{1}{4}. \end{cases}$$

Differentiating,

$$f_X(x) = \frac{2}{\sqrt{1-4x}}, \qquad 0 < x < \frac{1}{4}.$$

Thus,

$$\boxed{F_X(x) = \begin{cases} 0, & x < 0, \\ 1 - \sqrt{1-4x}, & 0 \le x \le \frac{1}{4}, \\ 1, & x > \frac{1}{4}, \end{cases}}$$

and

$$\boxed{f_X(x) = \begin{cases} \dfrac{2}{\sqrt{1-4x}}, & 0 < x < \dfrac{1}{4}, \\ 0, & \text{otherwise.} \end{cases}}$$

7. Let $X \sim \text{Binomial}(n, p)$. Define

$$Y = \frac{1}{X+1}.$$

(a) Show that

$$E\left(\frac{1}{X+1}\right) = \frac{1-(1-p)^{n+1}}{(n+1)p}.$$

Solution: By the binomial probability mass function,

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

Therefore,

$$E\left(\frac{1}{X+1}\right) = \sum_{k=0}^{n} \frac{1}{k+1} \binom{n}{k} p^k (1-p)^{n-k}.$$

Using

$$\frac{1}{k+1}\binom{n}{k} = \frac{1}{n+1}\binom{n+1}{k+1},$$

we obtain

$$E\left(\frac{1}{X+1}\right) = \frac{1}{n+1} \sum_{k=0}^{n} \binom{n+1}{k+1} p^k (1-p)^{n-k}.$$

Let $j = k+1$. Then

$$E\left(\frac{1}{X+1}\right) = \frac{1}{(n+1)p} \sum_{j=1}^{n+1} \binom{n+1}{j} p^j (1-p)^{n+1-j}.$$

By the binomial theorem,

$$\sum_{j=0}^{n+1} \binom{n+1}{j} p^j (1-p)^{n+1-j} = 1.$$

Removing the $j = 0$ term gives

$$\sum_{j=1}^{n+1} \binom{n+1}{j} p^j (1-p)^{n+1-j} = 1 - (1-p)^{n+1}.$$

Therefore,

$$\boxed{E\left(\frac{1}{X+1}\right) = \frac{1-(1-p)^{n+1}}{(n+1)p}.}$$

(b) Deduce a closed-form expression for

$$E\left(\frac{X}{X+1}\right).$$

Solution: Observe that

$$\frac{X}{X+1} = 1 - \frac{1}{X+1}.$$

Therefore,

$$\begin{aligned} E\left(\frac{X}{X+1}\right) &= 1 - E\left(\frac{1}{X+1}\right) \\ &= 1 - \frac{1-(1-p)^{n+1}}{(n+1)p}. \end{aligned}$$

Thus,

$$\boxed{E\left(\frac{X}{X+1}\right) = 1 - \frac{1-(1-p)^{n+1}}{(n+1)p}.}$$

CS 215: Task 05: Due: 04-Sep; time as announced on Piazza. No extensions.

- Your answers should be in your handwriting. Try to be neat and clear. If an LLM can read your handwriting, we might be able to.
- If you used any source (person or thing) explicitly state it. This is an individual assignment.

Teaching-team notes.

- When we say, 'derive' or 'show', simply put 2-3 important steps in the derivation in the space provided. Please attach extra paper for your records and for you to scan in the upload.

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0. Please write the honor code in your handwriting, with the understanding that it is true.

1. X and Y are independent random variables with MGFs $M_X(t)$ and $M_Y(t)$. Derive the MGF of

$$W = aX + bY + c.$$

Hence, find $E[W]$ and $\mathrm{Var}(W)$.

Teaching-team notes. Use only the independence property mentioned in the slides (Aug 24-Slide 13).

2. $X_1, X_2, X_3 \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$. Let $T = X_{(2)}$ denote their median, i.e., the middle value when $X_{(1)} \le X_{(2)} \le X_{(3)}$ are in order. What is the pdf of T?

3. Let

$$X \sim \text{Bin}(n, p), \qquad Y \sim \text{Bin}(m, p)$$

be independent.

(a) Show that the conditional distribution of X, given $X + Y = k$, (i) is hypergeometric, and (ii) identify its parameters.

Teaching-team notes. Notation: $Z \sim \text{Hypergeometric}(N, W, K)$ if Z is the number of successes in a sample of size W drawn without replacement from a population of N objects containing K successes.

(b) Explain in words, without further calculation, why the parameter p disappears after conditioning.

4. A random ordering is made of a population containing K successes among N objects. In that ordering, let X be the number of successes among the first n_1 positions and Y the number among the next n_2 positions, where $n_1 + n_2 \leq N$. Find $\text{Cov}(X, Y)$.

   (a) Explain, without any calculation, the sign of the covariance.

   (b) Find $E[X]$ and $E[Y]$.

   (c) Find $\text{Cov}(X, Y)$.

5. Let $X_1, X_2, \ldots, X_k$ be independent random variables such that

$$X_i \sim \text{Geom}(p_i), \qquad i = 1, \ldots, k,$$

where the geometric distribution is supported on $\{1, 2, 3, \ldots\}$, i.e.,

$$P(X_i = n) = (1 - p_i)^{n-1} p_i, \qquad 0 < p_i < 1.$$

Define

$$M = \min(X_1, X_2, \ldots, X_k).$$

(a) Determine the distribution of M and its parameter(s).

(b) Find the exact probability that variable j achieves the minimum, i.e.,

$$P(X_j = M).$$

6. Let $X \sim N(0, \sigma^2)$ and $Y = |X|$. Let $\Phi(x)$ and $\phi(x)$ be the CDF and PDF of a standard normal random variable (i.e., mean 0, variance 1).

   (a) Derive the CDF and PDF of Y in terms of $\Phi(x)$ and $\phi(x)$.

(b) Compute $E[Y]$.

7. In an election, each of 1000 voters independently votes for candidate A with probability 0.52. We want to determine the probability that A receives a strict (i.e. ties not permitted) majority of the votes.

   (a) Complete the one-liner `python` script to compute this probability.

```
from scipy.stats import ????
prob =  ????
print(f"{prob:.4f}")
```

   (b) Use the normal approximation, the table in the Appendix of the textbook, and appropriate approximations to hand-calculate the probability without using any electronic aid.

(c) Use the normal approximation and `python` to compute the probability. Write the one-liner for the code fragment.

(d) Are you able to obtain the probability correct to 2 decimal places? If so, what approximations did you use?

8. The annual salaries of physicians are normally distributed. It is known that (in some currency)

   - 25% earn less than 180,000, and
   - 30% earn more than 320,000.

(a) Find the mean (μ) and standard deviation (σ) of the salary distribution.

(b) Find the fraction earning between 260,000 and 300,000.

CS 215: Task 05: September 14, 2026

1. X and Y are independent random variables with MGFs $M_X(t)$ and $M_Y(t)$. Derive the MGF of

$$W = aX + bY + c.$$

Hence, find $E[W]$ and $\mathrm{Var}(W)$.

Teaching-team notes. Use only the independence property mentioned in the slides (Aug 24-Slide 13).

Solution: The MGF of W is

$$\begin{aligned} M_W(t) &= E[e^{tW}] \\ &= E[e^{t(aX+bY+c)}] \\ &= e^{ct}E[e^{atX}e^{btY}] \\ &= e^{ct}M_X(at)M_Y(bt), \end{aligned}$$

where independence of X and Y is used. Hence,

$$\boxed{M_W(t) = e^{ct}M_X(at)M_Y(bt)}.$$

Differentiating and evaluating at $t = 0$, using $M_X(0) = M_Y(0) = 1$,

$$M'_W(0) = aM'_X(0) + bM'_Y(0) + c,$$

so

$$\boxed{E[W] = aE[X] + bE[Y] + c}.$$

Similarly, using

$$\mathrm{Var}(W) = M''_W(0) - [M'_W(0)]^2,$$

we obtain

$$\boxed{\mathrm{Var}(W) = a^2\,\mathrm{Var}(X) + b^2\,\mathrm{Var}(Y)}.$$

2. $X_1, X_2, X_3 \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$. Let $T = X_{(2)}$ denote their median, i.e., the middle value when $X_{(1)} \le X_{(2)} \le X_{(3)}$ are in order. What is the pdf of T?

Solution: Let

$$F(t) = P(X_i \le t) = \Phi\left(\frac{t-\mu}{\sigma}\right),$$

where Φ is the standard normal CDF.

For the median to be at most t, at least two of the three observations must be at most t. Hence,

$$\begin{aligned} F_T(t) &= P(T \le t) \\ &= \binom{3}{2}F(t)^2[1 - F(t)] + F(t)^3 \\ &= 3F(t)^2 - 2F(t)^3. \end{aligned}$$

Thus,

$$\boxed{F_T(t) = 3F(t)^2 - 2F(t)^3}.$$

Differentiating,

$$f_T(t) = 6F(t)[1 - F(t)]f(t),$$

where

$$f(t) = \frac{1}{\sigma}\phi\left(\frac{t-\mu}{\sigma}\right).$$

Therefore,

$$\boxed{f_T(t) = \frac{6}{\sigma}\Phi\left(\frac{t-\mu}{\sigma}\right)\left[1 - \Phi\left(\frac{t-\mu}{\sigma}\right)\right]\phi\left(\frac{t-\mu}{\sigma}\right)}.$$

3. Let

$$X \sim \text{Bin}(n, p), \qquad Y \sim \text{Bin}(m, p)$$

be independent.

(a) Show that the conditional distribution of X, given $X + Y = k$, (i) is hypergeometric, and (ii) identify its parameters.

Teaching-team notes. Notation: $Z \sim \text{Hypergeometric}(N, W, K)$ if Z is the number of successes in a sample of size W drawn without replacement from a population of N objects containing K successes.

Solution: Note that, as mentioned in class, $X + Y \sim \text{Bin}(n + m, p)$. For x such that

$$\max(0, k - m) \le x \le \min(n, k),$$

we have

$$\begin{aligned} P(X = x \mid X + Y = k) &= \frac{P(X = x, Y = k - x)}{P(X + Y = k)} \\ &= \frac{\binom{n}{x}p^x(1-p)^{n-x}\binom{m}{k-x}p^{k-x}(1-p)^{m-k+x}}{\binom{n+m}{k}p^k(1-p)^{n+m-k}} \\ &= \frac{\binom{n}{x}\binom{m}{k-x}}{\binom{n+m}{k}}. \end{aligned}$$

Therefore, per notation, with $N = n + m, W = k, K = n$

$$\boxed{X \mid (X + Y = k) \sim \text{Hypergeometric}(n + m, k, n)}.$$

(b) Explain in words, without further calculation, why the parameter p disappears after conditioning.

Solution: Since $X_1, \ldots, X_n, Y_1, \ldots, Y_m$ (the underlying trials) are iid Bernoulli(p), every arrangement of successes and failures among the $n+m$ trials with exactly k successes has the same probability, $p^k(1-p)^{n+m-k}$, regardless of which k positions are the successes. So, conditional on there being k successes in total, all $\binom{n+m}{k}$ arrangements are equally likely; this is exactly the

same as randomly choosing which k of the $n+m$ trials are "black balls," with no dependence on p – equivalent to drawing without replacement from an urn.

4. A random ordering is made of a population containing K successes among N objects. In that ordering, let X be the number of successes among the first n_1 positions and Y the number among the next n_2 positions, where $n_1 + n_2 \le N$. Find $\mathrm{Cov}(X, Y)$.

   (a) Explain, without any calculation, the sign of the covariance.

   Solution: The covariance is **nonpositive** because sampling is without replacement: if the first portion contains more successes, fewer successes remain available for the second portion, and vice versa.

   (b) Find $E[X]$ and $E[Y]$.

   Solution: Let I_i be the indicator that the i-th position contains a success. As mentioned in class, each position has probability K/N of being a success. Therefore,

$$\boxed{E[X] = \frac{n_1 K}{N}}, \qquad \boxed{E[Y] = \frac{n_2 K}{N}}.$$

   (c) Find $\mathrm{Cov}(X, Y)$.

   Solution: Let I_i be the indicator that the i-th position contains a success. Then

$$X = \sum_{i=1}^{n_1} I_i, \qquad Y = \sum_{j=n_1+1}^{n_1+n_2} I_j.$$

   For two distinct positions $i \neq j$, $N > 1$,

$$P(I_i = 1, I_j = 1) = \frac{K}{N}\frac{K-1}{N-1}.$$

   Since

$$E[I_i] = E[I_j] = \frac{K}{N},$$

   we get

$$\begin{aligned} \mathrm{Cov}(I_i, I_j) &= E[I_i I_j] - E[I_i]E[I_j] \\ &= \frac{K(K-1)}{N(N-1)} - \frac{K^2}{N^2} \\ &= -\frac{K(N-K)}{N^2(N-1)}. \end{aligned}$$

   Every pair consisting of one position from the first portion and one from the second portion has this same covariance. There are $n_1 n_2$ such pairs. Hence,

$$\mathrm{Cov}(X, Y) = n_1 n_2 \,\mathrm{Cov}(I_i, I_j),$$

   so

$$\boxed{\mathrm{Cov}(X, Y) = -\frac{n_1 n_2 K(N-K)}{N^2(N-1)}}.$$

5. Let $X_1, X_2, \ldots, X_k$ be independent random variables such that

$$X_i \sim \text{Geom}(p_i), \qquad i = 1, \ldots, k,$$

where the geometric distribution is supported on $\{1, 2, 3, \ldots\}$, i.e.,

$$P(X_i = n) = (1 - p_i)^{n-1} p_i, \qquad 0 < p_i < 1.$$

Define

$$M = \min(X_1, X_2, \ldots, X_k).$$

(a) Determine the distribution of M and its parameter(s).

Solution: Let

$$q_i = 1 - p_i.$$

For $m = 1, 2, \ldots,$

$$\begin{aligned} P(M > m) &= P(X_1 > m, \ldots, X_k > m) \\ &= \prod_{i=1}^{k} P(X_i > m) \\ &= \prod_{i=1}^{k} q_i^m \\ &= \left(\prod_{i=1}^{k} q_i \right)^m. \end{aligned}$$

Hence,

$$\begin{aligned} P(M = m) &= P(M > m - 1) - P(M > m) \\ &= \left(\prod_{i=1}^{k} q_i \right)^{m-1} \left(1 - \prod_{i=1}^{k} q_i \right). \end{aligned}$$

Therefore,

$$\boxed{M \sim \text{Geom}\left(1 - \prod_{i=1}^{k} (1 - p_i) \right)}.$$

Thus, the parameter of M is

$$\boxed{p_M = 1 - \prod_{i=1}^{k} (1 - p_i)}.$$

(b) Find the exact probability that variable j achieves the minimum, i.e.,

$$P(X_j = M).$$

Solution: We have

$$\begin{aligned} P(X_j = M) &= \sum_{m=1}^{\infty} P(X_j = m,\ X_i \geq m \text{ for all } i \neq j) \\ &= \sum_{m=1}^{\infty} p_j q_j^{m-1} \prod_{i \neq j} q_i^{m-1} \\ &= p_j \sum_{m=1}^{\infty} \left(\prod_{i=1}^{k} q_i \right)^{m-1}. \end{aligned}$$

Using the geometric series,

$$\sum_{m=1}^{\infty} r^{m-1} = \frac{1}{1-r},$$

we obtain

$$\boxed{P(X_j = M) = \frac{p_j}{1 - \prod_{i=1}^{k}(1-p_i)}}.$$

This probability includes the possibility of a tie at the minimum.

6. Let $X \sim N(0, \sigma^2)$ and $Y = |X|$. Let $\Phi(x)$ and $\phi(x)$ be the CDF and PDF of a standard normal random variable (i.e., mean 0, variance 1).

   (a) Derive the CDF and PDF of Y in terms of $\Phi(x)$ and $\phi(x)$.

   Solution: For $y < 0$,

   $$F_Y(y) = P(Y \leq y) = 0.$$

   For $y \geq 0$,

   $$\begin{aligned} F_Y(y) &= P(|X| \leq y) \\ &= P(-y \leq X \leq y) \\ &= \Phi\left(\frac{y}{\sigma}\right) - \Phi\left(-\frac{y}{\sigma}\right) \\ &= 2\Phi\left(\frac{y}{\sigma}\right) - 1. \end{aligned}$$

   Hence,

   $$\boxed{F_Y(y) = \begin{cases} 0, & y < 0, \\ 2\Phi\left(\frac{y}{\sigma}\right) - 1, & y \geq 0. \end{cases}}$$

   Differentiating,

   $$\boxed{f_Y(y) = \frac{2}{\sigma}\phi\left(\frac{y}{\sigma}\right), \qquad y \geq 0,}$$

   and $f_Y(y) = 0$ for $y < 0$.

   (b) Compute $E[Y]$.

   Solution: To find the mean,

   $$E[Y] = E[|X|] = 2\int_0^{\infty} x \frac{1}{\sigma}\phi\left(\frac{x}{\sigma}\right) dx = \boxed{\sigma\sqrt{\frac{2}{\pi}}}.$$

7. In an election, each of 1000 voters independently votes for candidate A with probability 0.52. We want to determine the probability that A receives a strict (i.e. ties not permitted) majority of the votes.

   (a) Complete the one-liner `python` script to compute this probability.

   ```
   from scipy.stats import ????
   prob =  ????
   print(f"{prob:.4f}")
   ```

   Solution:

   ```
   from scipy.stats import binom
   prob =  1 - binom.cdf(500, 1000, 0.52)
   print(f"{prob:.4f}")
   ```

   Value comes out to 0.8914

   (b) Use the normal approximation, the table in the Appendix of the textbook, and appropriate approximations to hand-calculate the probability without using any electronic aid.

   Solution: Let $X \sim \text{Binomial}(n = 1000, p = 0.52)$ be the number of votes for candidate A.
   With an electronic aid, we have

   - Mean: $\mu = np = 1000 \times 0.52 = 520$
   - Variance: $\sigma^2 = np(1-p) = 1000 \times 0.52 \times 0.48 = 249.6 \implies \sigma \approx 15.80$

   Without an electronic aid, we approximate the variance to 250 which is slightly higher.
   A strict majority requires $X \geq 501$. Applying the normal approximation with continuity correction: With an electronic aid

$$\begin{aligned} P(X \geq 501) &\approx P(X \geq 500.5) \\ &= P\left(Z \geq \frac{500.5 - 520}{\sqrt{249.6}}\right) \\ &= P(Z \geq -1.234) \\ &= \Phi(1.234) \approx \mathbf{0.89} \end{aligned}$$

   Without an electronic aid

$$\begin{aligned} P(X \geq 501) &\approx P(X \geq 500.5) \\ &\approx P\left(Z \geq \frac{500.5 - 520}{\sqrt{250}}\right) \\ &\approx P(Z \geq -1.25) \\ &= \Phi(1.25) \approx \mathbf{0.89} \end{aligned}$$

   (c) Use the normal approximation and `python` to compute the probability. Write the one-liner for the code fragment.

   Solution:

   ```
   from scipy.stats import norm
   prob =  1 - norm.cdf(500.5, 1000*0.52, (1000*0.52*(1-0.52))**0.5)
   print(f"{prob:.4f}")
   ```

   Value comes to 0.8914

   (d) Are you able to obtain the probability correct to 2 decimal places? If so, what approximations did you use?

Solution: Yes. The normal approximation with continuity correction gives 0.89 to two decimal places. We also approximate the z-value using the normal table.

8. The annual salaries of physicians are normally distributed. It is known that (in some currency)

- 25% earn less than 180,000, and
- 30% earn more than 320,000.

(a) Find the mean (μ) and standard deviation (σ) of the salary distribution.

Solution: Let X denote the salary in thousands. From the given conditions:

$$P(X < 180) = 0.25 \implies \Phi\left(\frac{180-\mu}{\sigma}\right) = 0.25 \tag{1}$$

$$P(X > 320) = 0.30 \implies P(X \le 320) = 0.70 \implies \Phi\left(\frac{320-\mu}{\sigma}\right) = 0.70 \tag{2}$$

Using standard normal critical values:

- $z_{0.25} = -\Phi^{-1}(0.75) \approx -0.6745$
- $z_{0.70} = \Phi^{-1}(0.70) \approx 0.5244$

This yields the linear system:

$$\begin{aligned} 180 - \mu &= -0.6745\,\sigma \implies \mu - 0.6745\,\sigma = 180 \\ 320 - \mu &= 0.5244\,\sigma \implies \mu + 0.5244\,\sigma = 320 \end{aligned}$$

Subtracting the first equation from the second:

$$(0.5244 + 0.6745)\,\sigma = 320 - 180 \implies 1.1989\,\sigma = 140$$

$$\sigma = \frac{140}{1.1989} \approx 116.774 \implies \sigma \approx 116{,}774$$

Substituting σ back to find μ:

$$\mu = 180 + 0.6745(116.774) \approx 258.764 \implies \mu \approx 258{,}764$$

(b) Find the fraction earning between 260,000 and 300,000.

Solution: We evaluate $P(260 \le X \le 300)$:

$$z_1 = \frac{260 - 258.764}{116.774} = \frac{1.236}{116.774} \approx 0.0106$$

$$z_2 = \frac{300 - 258.764}{116.774} = \frac{41.236}{116.774} \approx 0.3531$$

Using the standard normal CDF Φ:

$$\begin{aligned} P(260 \le X \le 300) &= \Phi(0.3531) - \Phi(0.0106) \\ &\approx 0.6380 - 0.5042 \\ &= \mathbf{0.1338} \quad (\approx 13.38\%) \end{aligned}$$

CS 215: Mock Exam

Name:	ID:

Notes:

Teaching-team notes.

1. When in doubt, state any valid assumptions and proceed. Answers without reasoning do not qualify for full credit. Write all answers on this question paper.
2. Use the (non-printed) back of the paper for rough work. These will not be used to award credit.
3. Anything written outside the box will not be evaluated.
4. You may assume any result stated in provided lecture handouts as "known" i.e., not to be proved. A specific instruction may override this, so read carefully.

1. (6mark) For independent, normal $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$ and $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$, provide the mean and variance of the following if they are necessarily normal. Give one line reason (see the last note).
   A. $3X - 2$ B. $X + Y$ C. XY D. $|X|$ E. e^X F. $X - Y$

1. (6 marks) For independent, normal $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$ and $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$, provide the mean and variance of the following if they are necessarily normal. Give one line reason (see the last note in the instructions).

   A. $3X - 2$ B. $X + Y$ C. XY D. $|X|$ E. e^X F. $X - Y$

   Solution:

   - **$3X - 2$: Normal**. Any affine transformation of a normal random variable is normal:
   $$3X - 2 \sim N(3\mu_X - 2, 9\sigma_X^2).$$
   - **$X + Y$: Normal**. Since X and Y are independent normal random variables, their sum is normal:
   $$X + Y \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2).$$
   - **XY: Not necessarily normal**. The product of two independent normal random variables is generally not normal. First, for a case when it is normal, take X unit normal and Y=constant.

     For the intuition behind the counter example, take X,Y unit normal. For small values of X, the product is getting pushed to the origin. For occasionally large values of X, the product becomes very large. The distribution concentrates on the zero, and the "tails" are very large. This is not the behaviour of a normal where the tails drop off very quickly;(s.d. of 3 or above are hard to find.)

     Teaching-team notes. This is good enough for the exam – noting that the question as a whole is worth only 6 marks. (Sahil Deo's piazza post is cleaner.)

     There is actually a formal argument for the tail behaviour. You can do this by (a) noting that MGF of a unit normal is defined at every value of t. MGF of XY is undefined for mod(t) > 1.

     The constant case is in fact one of the rare cases when the product is normal, and again, here we are skirting on the boundary situation.
   - **$|X|$: Not normal**. For example, if $X \sim N(0, 1)$, then $|X|$ has a half-normal distribution, not a normal distribution. This is not normal as it doesn't take negative values.
   - **e^X: Not normal**. If X is normal, then e^X has a log-normal distribution. This is not normal as it doesn't take negative values.
   - **$X - Y$: Normal**. Since $-Y$ is normal and independent of X, their sum $X + (-Y) = X - Y$ is normal:
   $$X - Y \sim N(\mu_X - \mu_Y, \sigma_X^2 + \sigma_Y^2).$$

   Hence, the transformations that are necessarily normal are

   $$\boxed{3X - 2, \quad X + Y, \quad X - Y}.$$

CS 215: Task 06: Due: 7-Oct; time as announced on Piazza. No extensions.

- Your answers should be in your handwriting. Try to be neat and clear. If an LLM can read your handwriting, we might be able to.
- If you used any source (person or thing) explicitly state it. This is an individual assignment.

Teaching-team notes.

- For all questions in this task, explicitly write down the assumptions made.
- For all problems here, do not use electronic calculating aids since these will not be allowed in the exam. You may refer to the tables mentioned in the appendix.

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0. Please write the honor code in your handwriting, with the understanding that it is true.

1. The time taken by non-hostel resident students to travel to IIT Bombay has mean 42 minutes and standard deviation 12 minutes. Travel times on different days may be treated as independent.

   Suppose the travel times over 64 days are recorded. What is the probability that the average travel time is at least 39 minutes and at most 45 minutes?

2. The amount spent by a typical customer at Cafe92 has mean Rs. 180 and a standard deviation Rs. 90. Amounts spent by different customers may be treated as independent. On a particular evening, 100 customers visit Cafe92. What is the probability in percentage that the total revenue exceeds Rs. 20,000?

3. For each student in a class, independently, the probability of attending an optional review session is 0.62. A total of 400 students are eligible to attend. Using the Central Limit Theorem, approximate the probability that the number of students attending is between 240 and 260, inclusive.

4. The lifetime of an electronic chip has mean 500 hours and standard deviation 20 hours. Lifetimes of different chips may be treated as independent. The distribution of the lifetime of an individual component is not assumed to be normal.

   A student wishes to purchase n independently selected chips. Let $\overline{X}$ denote their average lifetime. Using the Central Limit Theorem, determine the smallest integer n for which

   $$P\left(|\overline{X} - 500| < 5\right) \geq 0.95.$$

   Teaching-team notes.

   $$P(|Z| \leq 1.96) \approx 0.95, \qquad Z \sim \mathcal{N}(0,1).$$

   You are expected to remember this number 1.96 during the exam, much the way you remember $\pi \approx 3.14$. It corresponds to the 95% confidence interval.

5. The weight of a packaged item has mean 100 grams and standard deviation 24 grams. Weights of different items may be treated as independent. A distributor is allowed to return a shipment back to the manufacturer if the average weight of the items in the shipment is more than c since the consumer gets more quantity for the same price.

   A distributor handles a random shipment of 144 items. Let $\overline{X}$ denote the average weight of the items.

   Using the Central Limit Theorem, find the value c such that approximately

   $$P(\overline{X} \leq c) = 0.975.$$

   Teaching-team notes. You may use

   $$P(Z \leq 1.96) = 0.975, \qquad Z \sim \mathcal{N}(0,1).$$

6. Suppose that, independently for each student, the probability that a student uses public transport to commute to campus is 0.40.

   A random sample of 600 students is chosen. Let $\widehat{P}$ denote the proportion of students in the sample who use public transport.

   Using the Central Limit Theorem for proportions, approximate

   $$P(0.36 \leq \widehat{P} \leq 0.44).$$

7. For each of the following cases, consider

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

For each case, determine whether the Central Limit Theorem as stated in class can be applied to approximate the distribution of $\overline{X}_n$ for large n. If it can, give the approximate distribution. If it cannot, state which assumption is missing. Write your answer starting with a phrase such as "Applies".

(a) The random variables $X_1, \ldots, X_n$ are i.i.d. with

$$E[X_i] = 5, \qquad \mathrm{Var}(X_i) = 9.$$

(b) The random variables $X_1, \ldots, X_n$ are i.i.d. with

$$E[X_i] = 5, \qquad E[X_i^2] = \infty.$$

(c) The random variables $X_1, \ldots, X_n$ are independent and satisfy

$$E[X_i] = 5, \qquad \mathrm{Var}(X_i) = 9,$$

but they are not assumed to have the same distribution.

(d) The random variables $X_1, \ldots, X_n$ are i.i.d. with

$$X_i \sim \mathcal{N}(5, 9).$$

In this case, also state whether the distribution obtained for $\overline{X}_n$ is approximate or exact.

8. Let $X_1, X_2, \ldots$ be i.i.d. random variables satisfying

$$E[X_i] = \mu, \qquad \mathrm{Var}(X_i) = \sigma^2,$$

and suppose

$$E[(X_i - \mu)^3] = \gamma\sigma^3,$$

where $\gamma \neq 0$.

For

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i,$$

define the standardized sample mean

$$Z_n = \frac{\sqrt{n}(\overline{X}_n - \mu)}{\sigma}.$$

(a) Show that

$$E[Z_n] = 0, \qquad \mathrm{Var}(Z_n) = 1.$$

(b) Show that

$$E[Z_n^3] = \frac{\gamma}{\sqrt{n}}.$$

(c) Compute $E[Z_n^3]$ for

$$n = 10, \qquad n = 200, \qquad n = 1000$$

and compare their magnitudes.

(d) Now take five independent batches, each containing 200 iid observations from the same distribution as X_i. Let $\overline{X}_{200}^{(j)}$ be the sample mean of batch j, and define

$$M = \frac{1}{5}\sum_{j=1}^{5} \overline{X}_{200}^{(j)}.$$

Show that M has the same mean, variance, and standardized third moment as a sample mean based on 1000 observations.

CS 215: Task 06: November 11, 2026

1. The time taken by non-hostel resident students to travel to IIT Bombay has mean 42 minutes and standard deviation 12 minutes. Travel times on different days may be treated as independent.

   Suppose the travel times over 64 days are recorded. What is the probability that the average travel time is at least 39 minutes and at most 45 minutes?

   Solution: No assumption is made about the distribution of the travel time on an individual day.

   $$\overline{X} = \frac{1}{64}\sum_{i=1}^{64} X_i$$

   denotes the average travel time. Assumption: Travel time on the 64 days are iid with mean and variance as stated $(42, 12^2)$.

   Using the Central Limit Theorem, we approximate

   $$P(39 \le \overline{X} \le 45).$$

   We have

   $$E[X_i] = 42, \qquad \mathrm{Var}(X_i) = 12^2.$$

   Therefore,

   $$E[\overline{X}] = 42, \qquad \mathrm{Var}(\overline{X}) = \frac{12^2}{64}.$$

   For $n = 64$, the Central Limit Theorem gives

   $$\overline{X} \approx \mathcal{N}\left(42, \frac{12^2}{64}\right).$$

   Thus,

   $$\mathrm{sd}(\overline{X}) = \frac{12}{8} = 1.5.$$

   Hence,

   $$\begin{aligned} P(39 \le \overline{X} \le 45) &\approx P\left(\frac{39-42}{1.5} \le Z \le \frac{45-42}{1.5}\right) \\ &= P(-2 \le Z \le 2). \end{aligned}$$

   Using the standard normal table,

   $$P(-2 \le Z \le 2) = 2\Phi(2) - 1 \approx 2(0.9772) - 1.$$

   Therefore,

   $$\boxed{P(39 \le \overline{X} \le 45) \approx 0.9544.}$$

2. The amount spent by a typical customer at Cafe92 has mean Rs. 180 and a standard deviation Rs. 90. Amounts spent by different customers may be treated as independent. On a particular evening, 100

customers visit Cafe92. What is the probability in percentage that the total revenue exceeds Rs. 20,000?

Solution: Let

$$T = \sum_{i=1}^{100} X_i$$

denote the total amount spent by these customers where

$$X_1, \ldots, X_{100} \text{ are assumed independent and identically distributed,}$$

with

$$E[X_i] = 180, \qquad \text{Var}(X_i) = 90^2.$$

Using the Central Limit Theorem, we approximate

$$P(T > 20000).$$

For each customer,

$$E[X_i] = 180, \qquad \text{Var}(X_i) = 90^2.$$

Therefore,

$$E[T] = 100(180) = 18000,$$

and, using independence,

$$\text{Var}(T) = 100(90^2).$$

Thus,

$$\text{sd}(T) = 90\sqrt{100} = 900.$$

By the Central Limit Theorem,

$$T \approx \mathcal{N}(18000, 900^2).$$

Hence,

$$\begin{aligned} P(T > 20000) &\approx P\left(Z > \frac{20000 - 18000}{900}\right) \\ &= P(Z > 2.22). \end{aligned}$$

Using the standard normal table,

$$P(Z > 2.22) = 1 - \Phi(2.22) \approx 1 - 0.9868.$$

Therefore,

$$\boxed{P(T > 20000) \approx 0.0132 \text{ i.e. } 1.32\%.}$$

3. For each student in a class, independently, the probability of attending an optional review session is 0.62. A total of 400 students are eligible to attend. Using the Central Limit Theorem, approximate the probability that the number of students attending is between 240 and 260, inclusive.

Solution:

Let W denote the number of students who attend the session.

Using a normal approximation with continuity correction, we want to approximate

$$P(240 \le W \le 260).$$

where

$$W \sim \text{Binomial}(400, 0.62).$$

Therefore,

$$E[W] = 400(0.62) = 248,$$

and

$$\text{Var}(W) = 400(0.62)(0.38) = 94.24.$$

Thus,

$$\text{sd}(W) = \sqrt{94.24} \approx 9.71.$$

Using continuity correction,

$$P(240 \le W \le 260) \approx P(239.5 < W < 260.5).$$

Therefore,

$$\begin{aligned} P(240 \le W \le 260) &\approx P\left(\frac{239.5 - 248}{9.71} < Z < \frac{260.5 - 248}{9.71}\right) \\ &\approx P(-0.88 < Z < 1.29). \end{aligned}$$

Using the standard normal table,

$$\begin{aligned} P(-0.88 < Z < 1.29) &= \Phi(1.29) - \Phi(-0.88) \\ &= 0.9015 - (1 - 0.8106) \\ &= 0.7121. \end{aligned}$$

Hence,

$$\boxed{P(240 \le W \le 260) \approx 0.712.}$$

4. The lifetime of an electronic chip has mean 500 hours and standard deviation 20 hours. Lifetimes of different chips may be treated as independent. The distribution of the lifetime of an individual component is not assumed to be normal.

   A student wishes to purchase n independently selected chips. Let $\overline{X}$ denote their average lifetime. Using the Central Limit Theorem, determine the smallest integer n for which

$$P\left(|\overline{X} - 500| < 5\right) \ge 0.95.$$

Teaching-team notes.

$$P(|Z| \le 1.96) \approx 0.95, \qquad Z \sim \mathcal{N}(0, 1).$$

You are expected to remember this number 1.96 during the exam, much the way you remember $\pi \approx 3.14$. It corresponds to the 95% confidence interval.

Solution: For a sample of size n,

$$E[\overline{X}] = 500$$

and

$$\mathrm{sd}(\overline{X}) = \frac{20}{\sqrt{n}}.$$

By the Central Limit Theorem,

$$\overline{X} \approx \mathcal{N}\left(500, \frac{20^2}{n}\right).$$

We require

$$P\left(|\overline{X} - 500| < 5\right) \geq 0.95.$$

Using

$$P(|Z| \leq 1.96) \approx 0.95,$$

we require

$$\frac{5}{20/\sqrt{n}} \geq 1.96.$$

Therefore,

$$\frac{\sqrt{n}}{4} \geq 1.96,$$

so

$$\sqrt{n} \geq 7.84.$$

Squaring,

$$n \geq 61.4656.$$

Since n must be an integer, the smallest possible value (using the CLT) is

$$\boxed{n = 62}.$$

5. The weight of a packaged item has mean 100 grams and standard deviation 24 grams. Weights of different items may be treated as independent. A distributor is allowed to return a shipment back to the manufacturer if the average weight of the items in the shipment is more than c since the consumer gets more quantity for the same price.

   A distributor handles a random shipment of 144 items. Let $\overline{X}$ denote the average weight of the items.

   Using the Central Limit Theorem, find the value c such that approximately

   $$P(\overline{X} \leq c) = 0.975.$$

Teaching-team notes. You may use

$$P(Z \leq 1.96) = 0.975, \qquad Z \sim \mathcal{N}(0, 1).$$

Solution: We have

$$E[\overline{X}] = 100$$

and

$$\mathrm{sd}(\overline{X}) = \frac{24}{\sqrt{144}} = 2.$$

By the Central Limit Theorem,

$$\overline{X} \approx \mathcal{N}(100, 2^2).$$

We want

$$P(\overline{X} \leq c) = 0.975.$$

Since

$$P(Z \leq 1.96) = 0.975,$$

we require

$$\frac{c - 100}{2} = 1.96.$$

Thus,

$$c - 100 = 3.92,$$

and hence

$$\boxed{c = 103.92 \text{ grams.}}$$

6. Suppose that, independently for each student, the probability that a student uses public transport to commute to campus is 0.40.

   A random sample of 600 students is chosen. Let $\widehat{P}$ denote the proportion of students in the sample who use public transport.

   Using the Central Limit Theorem for proportions, approximate

$$P(0.36 \leq \widehat{P} \leq 0.44).$$

Solution: Let X_i be an indicator variable that denotes 1 if student i uses public transport. Then

$$\widehat{P} = \sum_{1}^{600} X_i/600$$

We have

$$E[\widehat{P}] = 0.40$$

and

$$\text{Var}(\widehat{P}) = \frac{(0.40)(0.60)}{600} = 0.0004.$$

Therefore,

$$\text{sd}(\widehat{P}) = 0.02.$$

Using the Central Limit Theorem,

$$\widehat{P} \approx \mathcal{N}(0.40, 0.02^2).$$

Hence,

$$\begin{aligned} P(0.36 \leq \widehat{P} \leq 0.44) &\approx P\left(\frac{0.36 - 0.40}{0.02} \leq Z \leq \frac{0.44 - 0.40}{0.02}\right) \\ &= P(-2 \leq Z \leq 2). \end{aligned}$$

Using the standard normal table,

$$P(-2 \leq Z \leq 2) = 2\Phi(2) - 1 = 2(0.9772) - 1.$$

Thus,

$$\boxed{P(0.36 \leq \widehat{P} \leq 0.44) \approx 0.9544.}$$

7. For each of the following cases, consider

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

For each case, determine whether the Central Limit Theorem as stated in class can be applied to approximate the distribution of $\overline{X}_n$ for large n. If it can, give the approximate distribution. If it cannot, state which assumption is missing. Write your answer starting with a phrase such as "Applies".

(a) The random variables $X_1, \ldots, X_n$ are i.i.d. with

$$E[X_i] = 5, \qquad \text{Var}(X_i) = 9.$$

Solution:

Applies

The random variables are independent and identically distributed, and have finite mean and finite variance. Therefore, the Central Limit Theorem applies.

For large n,

$$\boxed{\overline{X}_n \approx \mathcal{N}\left(5, \frac{9}{n}\right).}$$

(b) The random variables $X_1, \ldots, X_n$ are i.i.d. with

$$E[X_i] = 5, \qquad E[X_i^2] = \infty.$$

Solution:

Does not apply.

Although the mean exists, the second moment is infinite, so the finite variance condition required by the version of the Central Limit Theorem used in class is not satisfied.

Hence, this version of the Central Limit Theorem cannot be applied.

(c) The random variables $X_1, \ldots, X_n$ are independent and satisfy

$$E[X_i] = 5, \qquad \text{Var}(X_i) = 9,$$

but they are not assumed to have the same distribution.

Solution: Cannot conclude authoritatively.

The random variables are independent and have the same mean and variance, but they are not assumed to be identically distributed.

Therefore, the i.i.d. version of the Central Limit Theorem stated in class cannot be directly applied.

Does not necessarily apply. Additional conditions that may allow non-identical random variables has not been stated in the question.

(d) The random variables $X_1, \ldots, X_n$ are i.i.d. with

$$X_i \sim \mathcal{N}(5, 9).$$

In this case, also state whether the distribution obtained for $\overline{X}_n$ is approximate or exact.

Solution:

Applies, not necessary.

Since the X_i are independent normal random variables, their average is itself normally distributed. Thus,

$$\boxed{\overline{X}_n \sim \mathcal{N}\left(5, \frac{9}{n}\right).}$$

This result is exact for every n; it does not require a large-n approximation.

8. Let $X_1, X_2, \ldots$ be i.i.d. random variables satisfying

$$E[X_i] = \mu, \qquad \mathrm{Var}(X_i) = \sigma^2,$$

and suppose

$$E[(X_i - \mu)^3] = \gamma\sigma^3,$$

where $\gamma \neq 0$.

For

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i,$$

define the standardized sample mean

$$Z_n = \frac{\sqrt{n}(\overline{X}_n - \mu)}{\sigma}.$$

(a) Show that

$$E[Z_n] = 0, \qquad \mathrm{Var}(Z_n) = 1.$$

Solution: Let

$$Y_i = X_i - \mu.$$

Then

$$E[Y_i] = 0, \qquad E[Y_i^2] = \sigma^2, \qquad E[Y_i^3] = \gamma\sigma^3.$$

Also,

$$Z_n = \frac{1}{\sigma\sqrt{n}}\sum_{i=1}^{n} Y_i.$$

Using linearity of expectation,

$$E[Z_n] = \frac{1}{\sigma\sqrt{n}}\sum_{i=1}^{n} E[Y_i] = 0.$$

Using independence,

$$\begin{aligned}
\mathrm{Var}(Z_n) &= \frac{1}{\sigma^2 n}\,\mathrm{Var}\left(\sum_{i=1}^{n} Y_i\right) \\
&= \frac{1}{\sigma^2 n}\sum_{i=1}^{n}\mathrm{Var}(Y_i) \\
&= \frac{n\sigma^2}{n\sigma^2} \\
&= 1.
\end{aligned}$$

Hence,

$$\boxed{E[Z_n] = 0, \qquad \mathrm{Var}(Z_n) = 1.}$$

(b) Show that

$$E[Z_n^3] = \frac{\gamma}{\sqrt{n}}.$$

Solution: We have

$$E[Z_n^3] = \frac{1}{\sigma^3 n^{3/2}} E\left[\left(\sum_{i=1}^{n} Y_i\right)^3\right].$$

When the cube is expanded, all mixed terms have expectation zero because the Y_i are independent and have mean zero. Therefore,

$$E\left[\left(\sum_{i=1}^{n} Y_i\right)^3\right] = \sum_{i=1}^{n} E[Y_i^3].$$

Thus,

$$\begin{aligned} E[Z_n^3] &= \frac{n\gamma\sigma^3}{\sigma^3 n^{3/2}} \\ &= \boxed{\frac{\gamma}{\sqrt{n}}}. \end{aligned}$$

(c) Compute $E[Z_n^3]$ for

$$n = 10, \qquad n = 200, \qquad n = 1000$$

and compare their magnitudes.

Solution: For the three sample sizes,

$$E[Z_{10}^3] = \frac{\gamma}{\sqrt{10}},$$

$$E[Z_{200}^3] = \frac{\gamma}{\sqrt{200}},$$

and

$$E[Z_{1000}^3] = \frac{\gamma}{\sqrt{1000}}.$$

Therefore,

$$\boxed{\left|E[Z_{1000}^3]\right| < \left|E[Z_{200}^3]\right| < \left|E[Z_{10}^3]\right|.}$$

The third standardized moment of a normal random variable is 0. Thus, this measure of asymmetry decreases towards zero as the sample size increases.

(d) Now take five independent batches, each containing 200 iid observations from the same distribution as X_i. Let $\overline{X}_{200}^{(j)}$ be the sample mean of batch j, and define

$$M = \frac{1}{5}\sum_{j=1}^{5} \overline{X}_{200}^{(j)}.$$

Show that M has the same mean, variance, and standardized third moment as a sample mean based on 1000 observations.

Solution: Write the observations in batch j as

$$X_{1j}, X_{2j}, \ldots, X_{200j}.$$

Then

$$\overline{X}_{200}^{(j)} = \frac{1}{200}\sum_{i=1}^{200} X_{ij}.$$

Hence,

$$\begin{aligned} M &= \frac{1}{5}\sum_{j=1}^{5} \overline{X}_{200}^{(j)} \\ &= \frac{1}{5}\sum_{j=1}^{5}\left(\frac{1}{200}\sum_{i=1}^{200} X_{ij}\right) \\ &= \frac{1}{1000}\sum_{j=1}^{5}\sum_{i=1}^{200} X_{ij}. \end{aligned}$$

Thus, M is simply the sample mean of all 1000 observations.

Therefore,

$$E[M] = \mu,$$

and

$$\mathrm{Var}(M) = \frac{\sigma^2}{1000}.$$

If

$$Z_M = \frac{\sqrt{1000}(M-\mu)}{\sigma},$$

then, by the result of part (b),

$$E[Z_M^3] = \boxed{\frac{\gamma}{\sqrt{1000}}}.$$

Thus, averaging five independent sample means based on 200 observations each gives the same mean, variance, and standardized third moment as taking one sample mean based on all 1000 observations.

CS 215: Task 07: Due: 15-Oct; time as announced on Piazza. No extensions.

- Your answers should be in your handwriting. Try to be neat and clear. If an LLM can read your handwriting, we might be able to.
- If you used any source (person or thing) explicitly state it. This is an individual task.

Teaching-team notes.

- For all questions in this task, explicitly write down the assumptions made.
- For all problems here, do not use electronic calculating aids since these will not be allowed in the exam. You may refer to the tables mentioned in the appendix of the textbook (5th edition).

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0. Please write the honor code in your handwriting, with the understanding that it is true.

1. Fill volumes of a bottling machine are $\mathcal{N}(\mu, \sigma^2)$ with $\sigma = 4$ ml. A random sample of $n = 16$ bottles gives $\bar{x} = 502$ ml.
   (a) If samples of 16 bottles were repeatedly drawn in this way, find an interval around the observed sample mean 502 ml such that the corresponding interval would contain μ in approximately 95% of repeated samples.

   (b) With the conditions as in part (a), find the smallest sample size for which the margin of error (half-width of the interval) is at most 1 ml.

   (c) How many bottles should we sample if we want the error to be at most 0.5 ml instead?

(d) Suppose the interval obtained in part (a) is written as

$$502 \pm \delta.$$

If a random sample of 100 individual bottles is taken, would you expect about 95 of them to have fill volumes within

$$502 \pm \delta?$$

Explain your answer.

2. Your supervisor asks you to estimate the mean running time of a piece of code. You run the program independently, and obtain the following running times (in ms):

$$12,\ 15,\ 11,\ 14,\ 13,\ 16,\ 10,\ 14,\ 12.$$

Assume that the running times are normally distributed.

(a) How would you report the mean running time to your supervisor? Use a 95% confidence level.

(b) Your assistant wrongly uses 1.96 in place of the t critical value. Compute the assistant's reported interval and explain the consequence for its actual coverage.

(c) A second assistant constructed a 99% confidence interval. Compare its width to the one in part (a).

3. Let $Z \sim \mathcal{N}(0,1)$.

(a) Show that

$$M_{Z^2}(t) = E\left[e^{tZ^2}\right] = (1-2t)^{-1/2}, \qquad t < \frac{1}{2}.$$

(b) Let $Z_1, \ldots, Z_k$ be i.i.d. $\mathcal{N}(0,1)$ and $W = \sum_{i=1}^{k} Z_i^2$. Find $M_W(t)$.

(c) Using $M_W(t)$, show that $E[W] = k$ and $\text{Var}(W) = 2k$.

(d) If $W_1 \sim \chi^2_a$ and $W_2 \sim \chi^2_b$ are independent, what is the distribution of $W_1 + W_2$? Justify your answer.

4. Let $X_1, \ldots, X_{10}$ be i.i.d. $\mathcal{N}(\mu, \sigma^2)$ and let S^2 be the sample variance.

(a) Find $E[S^2]$, $\text{Var}(S^2)$ and the standard deviation of S^2 expressed as a multiple of σ^2.

(b) Find constants c_1 and c_2 such that

$$P(S^2 < c_1\sigma^2) = 0.05, \qquad P(S^2 > c_2\sigma^2) = 0.05.$$

5. A random sample of $n = 11$ resistors from a normal population gives the sample standard deviation $s = 0.8\ \Omega$.

   (a) Construct a 95% confidence interval for σ^2.

   (b) Construct a 95% confidence interval for σ.

(c) Explain why the interval in part (a) is not symmetric about s^2.

(d) The specification requires $\sigma \le 1.5\ \Omega$. Based on your interval, what can you conclude at the 95% confidence level?

6. Let $X_1, \dots, X_5$ be i.i.d. $\mathcal{N}(0, \sigma^2)$ and define

$$Y = c\,\frac{X_1 + X_2}{\sqrt{X_3^2 + X_4^2 + X_5^2}}.$$

(a) Find the constant $c > 0$ for which Y has a student's t distribution, and state its degrees of freedom. c should not depend on σ.

(b) State which independence property you used in your construction.

CS 215: Task 07: Due: 15-Oct; time as announced on Piazza. No extensions.

1. Fill volumes of a bottling machine are $\mathcal{N}(\mu, \sigma^2)$ with $\sigma = 4$ ml. A random sample of $n = 16$ bottles gives $\bar{x} = 502$ ml.

   (a) If samples of 16 bottles were repeatedly drawn in this way, find an interval around the observed sample mean 502 ml such that the corresponding interval would contain μ in approximately 95% of repeated samples.

   Solution: Assume the bottle fill volumes are independent and identically distributed (i.i.d.) normal random variables, with known population standard deviation $\sigma = 4$ ml. Since the population standard deviation is known, the 95% confidence interval for μ is

   $$\bar{x} \pm z_{0.025}\frac{\sigma}{\sqrt{n}}.$$

   Substituting $\bar{x} = 502$, $z_{0.025} = 1.96$, $\sigma = 4$, and $n = 16$,

   $$502 \pm 1.96\frac{4}{\sqrt{16}} = 502 \pm 1.96.$$

   Thus, the required interval is

   $$\boxed{(500.04,\ 503.96) \text{ ml.}}$$

   In repeated sampling, approximately 95% of intervals constructed by this procedure would contain the fixed population mean μ.

   (b) With the conditions as in part (a), find the smallest sample size for which the margin of error (half-width of the interval) is at most 1 ml.

   Solution: For a known population standard deviation, the margin of error is

   $$E = z_{0.025}\frac{\sigma}{\sqrt{n}}.$$

   We require $E \leq 1$ ml. Therefore,

   $$1.96\frac{4}{\sqrt{n}} \leq 1 \quad \Longrightarrow \quad \sqrt{n} \geq 7.84 \quad \Longrightarrow \quad n \geq 7.84^2 = 61.4656.$$

   The sample size must be an integer, and rounding down would violate the required bound. Hence,

   $$\boxed{n = 62.}$$

   (c) How many bottles should we sample if we want the error to be at most 0.5 ml instead?

   Solution: The margin of error is $E = 1.96(4)/\sqrt{n}$. Requiring $E \leq 0.5$ gives

   $$1.96\frac{4}{\sqrt{n}} \leq 0.5 \quad \Longrightarrow \quad n \geq \left(\frac{1.96(4)}{0.5}\right)^2 = 245.8624.$$

   Rounding up to the next integer,

   $$\boxed{n = 246.}$$

   This is four times the approximate sample size needed for a 1 ml margin of error, because halving the margin of error requires multiplying the sample size by $2^2 = 4$.

(d) Suppose the interval obtained in part (a) is written as

$$502 \pm \delta.$$

If a random sample of 100 individual bottles is taken, would you expect about 95 of them to have fill volumes within

$$502 \pm \delta?$$

Explain your answer.

Solution: No. The interval $502 \pm \delta$ is a confidence interval for the population mean μ; it describes uncertainty in the estimate of the mean, not the spread of individual bottle fills. Individual fills have standard deviation 4 ml.

For an individual bottle $X \sim \mathcal{N}(\mu, 4^2)$, the central 95% of the population lies approximately within

$$\mu \pm 1.96(4).$$

Thus, the interval containing approximately 95% of individual bottle volumes is

$$\boxed{(\mu - 7.84,\ \mu + 7.84) \text{ ml.}}$$

The distinction is that the standard error of the sample mean is $\sigma/\sqrt{n}$, whereas the standard deviation of an individual observation is σ.

2. Your supervisor asks you to estimate the mean running time of a piece of code. You run the program independently, and obtain the following running times (in ms):

$$12,\ 15,\ 11,\ 14,\ 13,\ 16,\ 10,\ 14,\ 12.$$

Assume that the running times are normally distributed.

(a) How would you report the mean running time to your supervisor? Use a 95% confidence level.

Solution: Assume the nine running times are an independent sample from a normal population. Since the population standard deviation is unknown, use a t-interval with $n - 1 = 8$ degrees of freedom.

The sample mean is

$$\bar{x} = \frac{12 + 15 + 11 + 14 + 13 + 16 + 10 + 14 + 12}{9} = \frac{117}{9} = 13 \text{ ms.}$$

The deviations from the mean are

$$-1,\ 2,\ -2,\ 1,\ 0,\ 3,\ -3,\ 1,\ -1,$$

whose squared values sum to 30. Hence,

$$s^2 = \frac{30}{9-1} = 3.75, \qquad s = \sqrt{3.75} \approx 1.9365, \qquad \frac{s}{\sqrt{9}} \approx 0.6455.$$

Using $t_{0.025,8} = 2.306$, the 95% confidence interval is

$$\bar{x} \pm t_{0.025,8}\frac{s}{\sqrt{n}} = 13 \pm 2.306(0.6455) = 13 \pm 1.489.$$

Therefore,

$$\boxed{(11.51,\ 14.49) \text{ ms.}}$$

(b) Your assistant wrongly uses 1.96 in place of the t critical value. Compute the assistant's reported interval and explain the consequence for its actual coverage.

Solution: Using 1.96 in place of the appropriate t-critical value gives

$$13 \pm 1.96(0.6455) = 13 \pm 1.265,$$

so the assistant reports

$$\boxed{(11.73,\ 14.27) \text{ ms.}}$$

This interval is narrower than the correct 95% t-interval. The reason is that the population standard deviation is unknown and has been estimated by s; the t-distribution accounts for the additional uncertainty in this estimate and therefore has a larger critical value than the standard normal distribution. Using 1.96 here consequently gives actual coverage below 95%.

(c) A second assistant constructed a 99% confidence interval. Compare its width to the one in part (a).

Solution: For a 99% confidence interval with 8 degrees of freedom, use $t_{0.005,8} = 3.355$. The standard error is unchanged from part (a), so

$$13 \pm 3.355(0.6455) = 13 \pm 2.166.$$

Thus, the interval is

$$\boxed{(10.83,\ 15.17) \text{ ms.}}$$

It is wider than the 95% interval because a higher confidence level requires a larger critical value, while the sample size and standard error remain the same.

3. Let $Z \sim \mathcal{N}(0,1)$.

(a) Show that

$$M_{Z^2}(t) = E\left[e^{tZ^2}\right] = (1-2t)^{-1/2}, \qquad t < \frac{1}{2}.$$

Solution: The standard normal density is

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}.$$

Therefore,

$$\begin{aligned} M_{Z^2}(t) &= E[e^{tZ^2}] \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tz^2} e^{-z^2/2}\, dz \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(1-2t)z^2/2}\, dz. \end{aligned}$$

For this integral to be finite, we need $1 - 2t > 0$, or $t < 1/2$. Let $u = \sqrt{1-2t}\, z$. Then $dz = du/\sqrt{1-2t}$, and

$$M_{Z^2}(t) = \frac{1}{\sqrt{1-2t}} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-u^2/2}\, du = (1-2t)^{-1/2}.$$

Hence

$$\boxed{M_{Z^2}(t) = (1-2t)^{-1/2}, \qquad t < \frac{1}{2}.}$$

(b) Let $Z_1, \ldots, Z_k$ be i.i.d. $\mathcal{N}(0,1)$ and $W = \sum_{i=1}^{k} Z_i^2$. Find $M_W(t)$.

Solution: Since the Z_i's are independent, the random variables Z_i^2 are also independent. The MGF of a sum of independent random variables is the product of their MGFs. Thus,

$$M_W(t) = E\left[e^{t\sum_{i=1}^{k} Z_i^2}\right] = \prod_{i=1}^{k} E[e^{tZ_i^2}]$$
$$= \prod_{i=1}^{k} (1-2t)^{-1/2} = (1-2t)^{-k/2}, \qquad t < \frac{1}{2}.$$

Therefore,

$$\boxed{M_W(t) = (1-2t)^{-k/2}, \qquad t < \frac{1}{2}.}$$

(c) Using $M_W(t)$, show that $E[W] = k$ and $\mathrm{Var}(W) = 2k$.

Solution: From part (b),

$$M_W(t) = (1-2t)^{-k/2}.$$

Differentiating,

$$M_W'(t) = k(1-2t)^{-k/2-1},$$

and

$$M_W''(t) = k(k+2)(1-2t)^{-k/2-2}.$$

The first two moments are obtained by evaluating these derivatives at $t = 0$:

$$E[W] = M_W'(0) = k, \qquad E[W^2] = M_W''(0) = k(k+2).$$

Consequently,

$$\mathrm{Var}(W) = E[W^2] - (E[W])^2 = k(k+2) - k^2 = 2k.$$

Hence,

$$\boxed{E[W] = k, \qquad \mathrm{Var}(W) = 2k.}$$

(d) If $W_1 \sim \chi_a^2$ and $W_2 \sim \chi_b^2$ are independent, what is the distribution of $W_1 + W_2$? Justify your answer.

Solution: Because W_1 and W_2 are independent, the MGF of their sum is the product of their MGFs. Therefore,

$$M_{W_1+W_2}(t) = M_{W_1}(t) M_{W_2}(t)$$
$$= (1-2t)^{-a/2}(1-2t)^{-b/2}$$
$$= (1-2t)^{-(a+b)/2}.$$

This is the MGF of a chi-square random variable with $a + b$ degrees of freedom. Since MGFs uniquely determine distributions when they exist in a neighbourhood of zero,

$$\boxed{W_1 + W_2 \sim \chi_{a+b}^2.}$$

4. Let $X_1, \ldots, X_{10}$ be i.i.d. $\mathcal{N}(\mu, \sigma^2)$ and let S^2 be the sample variance.

(a) Find $E[S^2]$, $\mathrm{Var}(S^2)$ and the standard deviation of S^2 expressed as a multiple of σ^2.

Solution: Here $n = 10$. By the stated chi-square result,

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1},$$

so we can write $S^2 = (\sigma^2/9)W$, where $W \sim \chi^2_9$. A chi-square variable with 9 degrees of freedom has mean 9 and variance 18. Hence,

$$E[S^2] = \frac{\sigma^2}{9}E[W] = \frac{\sigma^2}{9}(9) = \sigma^2,$$

and

$$\mathrm{Var}(S^2) = \frac{\sigma^4}{9^2}\,\mathrm{Var}(W) = \frac{18\sigma^4}{81} = \frac{2\sigma^4}{9}.$$

Taking the square root,

$$\boxed{\mathrm{sd}(S^2) = \frac{\sqrt{2}}{3}\sigma^2 \approx 0.471\,\sigma^2.}$$

(b) Find constants c_1 and c_2 such that

$$P(S^2 < c_1\sigma^2) = 0.05, \qquad P(S^2 > c_2\sigma^2) = 0.05.$$

Solution: Let $W = 9S^2/\sigma^2$, so $W \sim \chi^2_9$. For the upper-tail probability,

$$P(S^2 > c_2\sigma^2) = P(W > 9c_2) = 0.05.$$

By the given chi-square critical value, $9c_2 = \chi^2_{0.05,9} = 16.919$, giving

$$c_2 = \frac{16.919}{9} \approx 1.880.$$

Similarly,

$$P(S^2 < c_1\sigma^2) = P(W < 9c_1) = 0.05.$$

Since $\chi^2_{0.95,9} = 3.325$ is the lower 5th percentile under the stated upper-tail convention,

$$9c_1 = 3.325 \quad\Longrightarrow\quad c_1 = \frac{3.325}{9} \approx 0.369.$$

Thus,

$$\boxed{c_1 \approx 0.369, \qquad c_2 \approx 1.880.}$$

5. A random sample of $n = 11$ resistors from a normal population gives the sample standard deviation $s = 0.8\ \Omega$.

(a) Construct a 95% confidence interval for σ^2.

Solution: Assume the resistor resistances are an i.i.d. sample from a normal population. For a normal population,

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}.$$

Here $n = 11$, $s = 0.8\ \Omega$, and $(n-1)s^2 = 10(0.8)^2 = 6.4$. Using the supplied critical values,

$$P\left(3.247 \le \frac{10S^2}{\sigma^2} \le 20.483\right) = 0.95.$$

To isolate σ^2, invert the inequalities. Since all quantities are positive, inversion reverses the order:

$$P\left(\frac{6.4}{20.483} \le \sigma^2 \le \frac{6.4}{3.247}\right) = 0.95.$$

The 95% confidence interval is therefore

$$\boxed{(0.312,\ 1.971)\ \Omega^2.}$$

(b) Construct a 95% confidence interval for σ.

Solution: Since $\sigma > 0$, take the square root of each endpoint of the variance interval from part (a):

$$\left(\sqrt{0.312},\ \sqrt{1.971}\right) \approx \boxed{(0.559,\ 1.404)\ \Omega.}$$

(c) Explain why the interval in part (a) is not symmetric about s^2.

Solution: The interval is obtained by transforming two chi-square critical values. The chi-square distribution is right-skewed, so its lower and upper critical values are not equally spaced around its mean (or around $n-1$). Moreover, solving the pivotal inequality for σ^2 puts the larger chi-square critical value in the denominator of the lower endpoint and the smaller one in the denominator of the upper endpoint. The resulting interval is therefore not symmetric about $s^2 = 0.64\ \Omega^2$.

(d) The specification requires $\sigma \le 1.5\ \Omega$. Based on your interval, what can you conclude at the 95% confidence level?

Solution: The 95% confidence interval for σ is (0.559, 1.404) Ω. Its upper endpoint is below the specification limit 1.5 Ω. Thus the entire confidence interval lies within the acceptable range, and we can conclude at the 95% confidence level that the population standard deviation satisfies

$$\boxed{\sigma < 1.5\ \Omega.}$$

6. Let $X_1, \ldots, X_5$ be i.i.d. $\mathcal{N}(0, \sigma^2)$ and define

$$Y = c\,\frac{X_1 + X_2}{\sqrt{X_3^2 + X_4^2 + X_5^2}}.$$

(a) Find the constant $c > 0$ for which Y has a student's t distribution, and state its degrees of freedom. c should not depend on σ.

Solution: The numerator is normal:

$$X_1 + X_2 \sim \mathcal{N}(0, 2\sigma^2), \qquad Z = \frac{X_1 + X_2}{\sigma\sqrt{2}} \sim \mathcal{N}(0,1).$$

For the denominator, each X_i/σ is standard normal, so

$$V = \frac{X_3^2 + X_4^2 + X_5^2}{\sigma^2} \sim \chi^2_3.$$

By the definition of the Student t-distribution, if $Z \sim \mathcal{N}(0,1)$, $V \sim \chi^2_\nu$, and Z, V are independent, then $Z/\sqrt{V/\nu} \sim t_\nu$. Here $\nu = 3$, and

$$\begin{aligned}\frac{Z}{\sqrt{V/3}} &= \frac{X_1 + X_2}{\sigma\sqrt{2}}\sqrt{\frac{3\sigma^2}{X_3^2 + X_4^2 + X_5^2}} \\ &= \sqrt{\frac{3}{2}}\,\frac{X_1 + X_2}{\sqrt{X_3^2 + X_4^2 + X_5^2}}.\end{aligned}$$

Comparing this with the definition of Y, we obtain

$$\boxed{c = \sqrt{\frac{3}{2}} \approx 1.225, \qquad Y \sim t_3.}$$

The factor σ cancels, as required.

(b) State which independence property you used in your construction.

Solution: The variable Z is a function only of X_1, X_2, while V is a function only of X_3, X_4, X_5. These two groups of observations are independent because all five X_i's are independent. Functions of independent groups of random variables remain independent. Therefore,

$$\boxed{Z \text{ and } V \text{ are independent.}}$$

CS 215: Task 08: Due: 30-Oct; time 7:00 AM

- Your answers should be in your handwriting. Try to be neat and clear. If an LLM can read your handwriting, we might be able to.
- If you used any source (person or thing) explicitly state it. This is an individual task.

Teaching-team notes.

- For all questions in this task, explicitly write down the assumptions made.
- For all problems here, do not use electronic calculating aids since these will not be allowed in the exam. You may refer to the tables mentioned in the appendix of the textbook (5th edition).

---

0. Please write the honor code in your handwriting, with the understanding that it is true.

1. In class, we derived the maximum likelihood estimator (MLE) for a normal distribution by first writing the probability density for a single observation x_i, then constructing the likelihood of the entire dataset using independence, and finally maximizing the log-likelihood.

   We now apply the same approach to a different probability model.

   Teaching-team notes. First, review the example in the book on the Bernoulli case. We are going from scalars to vectors in this question.

   In a population, a gene has two alleles, A [dominant] and a [recessive]. Every indvidual gets two copies of this gene (one from each parent) so there are three possible combinations (genotypes): AA, Aa, and aa. It is easy to see that they occur with probabilities

   $$p^2, \qquad 2p(1-p), \qquad (1-p)^2,$$

   respectively, where p is the unknown population frequency of the allele A.

   Suppose we independently sample N individuals. We observe counts of n_1 for AA, n_2 for Aa, and n_3 for aa (where $n_1 + n_2 + n_3 = N$).

   For each individual i, represent the observed genotype by a one-hot vector

   $$x_i = (x_{i1}, x_{i2}, x_{i3}),$$

   where $x_i = (1, 0, 0)$ denotes AA, $x_i = (0, 1, 0)$ denotes Aa, and $x_i = (0, 0, 1)$ denotes aa.

   (a) Write an expression for the probability $P(X_i = x_i; p)$ of a single observation, in terms of the components of x_i and the parameter p.

(b) Using the independence of the observations, construct the likelihood function $L(p)$ for the entire dataset $x_1, \ldots, x_N$.

(c) Derive the maximum likelihood estimator $\hat{p}$. Express your answer in terms of the numbers n_1, n_2, n_3 of individuals with genotypes AA, Aa, aa, respectively.

2. Let $X_1, \ldots, X_n$ be independent observations from a uniform distribution $U(0, \theta)$, where $\theta > 0$ is unknown.

(a) Write the likelihood function $L(\theta)$, explicitly stating the values of θ for which it is nonzero.

(b) Derive the maximum likelihood estimator of θ.

(c) Is the estimator unbiased? Justify your answer.

3. The returns on financial assets for (volatile) cryptocurrency are sometimes modeled using a Laplace distribution, which has heavier tails (extreme jumps are more common) than a Normal distribution.

   Suppose $X_1, \ldots, X_n$ are independent observations from a Laplace distribution with probability density function

$$f(x; \mu, b) = \frac{1}{2b} \exp\left(-\frac{|x - \mu|}{b}\right),$$

   where μ is an unknown location parameter and $b > 0$ is a known scale parameter.

   (a) Write down the likelihood and log-likelihood functions for the observed data $x_1, \ldots, x_n$.

   (b) Derive the maximum likelihood estimator of μ.

   (c) Compare this estimator with the MLE of the mean of a Normal distribution. Which estimator is less sensitive to extreme observations, and why?

4. A sample of 20 cigarettes is tested to determine nicotine content, and the average value observed is 1.2 mg. Suppose that the population variance is not known in advance, and the sample variance is 0.04.

   Assuming a normal population, compute a 99% two-sided confidence interval for the mean nicotine content of a cigarette.

5. A random sample of 1200 engineers includes 300 Hispanic American engineers, 400 African American engineers. Assume no overlap.

   Determine 90% confidence intervals

   (a) For the proportion of all engineers who are Hispanic American or African American.

   (b) The excess in the proportion of African Americans and proportion of Hispanic Americans

6. It is claimed that a certain type of bipolar transistor has a mean current gain of at least 210. A sample of these transistors has sample mean current gain 200 and sample standard deviation 35.

   Assuming that the population is normal, determine whether the claim should be rejected at the 5% level of significance if

   (a) the sample size is 25;

   (b) the sample size is 64.

CS 215: Task 08: Due: 30-Oct; 7:00 AM

1. In class, we derived the maximum likelihood estimator (MLE) for a normal distribution by first writing the probability density for a single observation x_i, then constructing the likelihood of the entire dataset using independence, and finally maximizing the log-likelihood.

   We now apply the same approach to a different probability model.

   Teaching-team notes. First, review the example in the book on the Bernoulli case. We are going from scalars to vectors in this question.

   In a population, a gene has two alleles, A [dominant] and a [recessive]. Every indvidual gets two copies of this gene (one from each parent) so there are three possible combinations (genotypes): AA, Aa, and aa. It is easy to see that they occur with probabilities

   $$p^2, \qquad 2p(1-p), \qquad (1-p)^2,$$

   respectively, where p is the unknown population frequency of the allele A.

   Suppose we independently sample N individuals. We observe counts of n_1 for AA, n_2 for Aa, and n_3 for aa (where $n_1 + n_2 + n_3 = N$).

   For each individual i, represent the observed genotype by a one-hot vector

   $$x_i = (x_{i1}, x_{i2}, x_{i3}),$$

   where $x_i = (1, 0, 0)$ denotes AA, $x_i = (0, 1, 0)$ denotes Aa, and $x_i = (0, 0, 1)$ denotes aa.

   (a) Write an expression for the probability $P(X_i = x_i; p)$ of a single observation, in terms of the components of x_i and the parameter p.

   Solution: For a single observation,

   $$\boxed{P(X_i = x_i; p) = (p^2)^{x_{i1}}[2p(1-p)]^{x_{i2}}[(1-p)^2]^{x_{i3}}}$$

   (b) Using the independence of the observations, construct the likelihood function $L(p)$ for the entire dataset $x_1, \ldots, x_N$.

   Solution: From independence we write

   $$L(p) = \prod_{i=1}^{N} P(X_i = x_i; p).$$

   (c) Derive the maximum likelihood estimator $\hat{p}$. Express your answer in terms of the numbers n_1, n_2, n_3 of individuals with genotypes AA, Aa, aa, respectively.

   Solution:

   1. **Define the likelihood function:**
      This is a multinomial distribution where the probabilities are constrained by a single parameter p.
      $$L(p) \propto (p^2)^{n_1}(2p(1-p))^{n_2}((1-p)^2)^{n_3}$$

We can drop the multinomial coefficient as it does not depend on p.

$$L(p) \propto p^{2n_1} \cdot p^{n_2}(1-p)^{n_2} \cdot (1-p)^{2n_3} = p^{2n_1+n_2}(1-p)^{n_2+2n_3}$$

2. **Take the log-likelihood:**

$$\ln L(p) = (2n_1 + n_2)\ln p + (n_2 + 2n_3)\ln(1-p) + \text{constant}$$

where by constant we indicate terms not related to the minimization.

3. **Differentiate and set to zero:**

$$\frac{d \ln L(p)}{dp} = \frac{2n_1 + n_2}{p} - \frac{n_2 + 2n_3}{1-p} = 0$$

4. **Solve for p:**

$$\begin{aligned}
(1-p)(2n_1 + n_2) &= p(n_2 + 2n_3) \\
2n_1 + n_2 - p(2n_1 + n_2) &= pn_2 + 2pn_3 \\
2n_1 + n_2 &= p(2n_1 + 2n_2 + 2n_3) \\
2n_1 + n_2 &= p(2N) \\
\hat{p}_{MLE} &= \frac{2n_1 + n_2}{2N}
\end{aligned}$$

2. Let $X_1, \ldots, X_n$ be independent observations from a uniform distribution $U(0, \theta)$, where $\theta > 0$ is unknown.
   (a) Write the likelihood function $L(\theta)$, explicitly stating the values of θ for which it is nonzero.
   (b) Derive the maximum likelihood estimator of θ.
   (c) Is the estimator unbiased? Justify your answer.

   Solution:

1. **Define the likelihood function:**
   The PDF of $U(0, \theta)$ is $\frac{1}{\theta}$ for $0 \leq x_i \leq \theta$, and 0 otherwise.

$$L(\theta) = \prod_{i=1}^{n} \frac{1}{\theta} = \frac{1}{\theta^n}$$

Crucial step: This likelihood is only valid if *every* observed x_i is less than or equal to θ. If even a single $x_i > \theta$, the likelihood drops to 0. We express this using the maximum order statistic $X_{(n)} = \max(x_1, \ldots, x_n)$.

$$L(\theta) = \begin{cases} \frac{1}{\theta^n} & \text{if } \theta \geq X_{(n)} \\ 0 & \text{if } \theta < X_{(n)} \end{cases}$$

2. **Maximize the likelihood:**
   Calculus fails here because the function is monotonically decreasing for $\theta > 0$. If we take the derivative of θ^{-n}, we get $-n\theta^{-n-1}$, which is never zero.

   Instead, we maximize $L(\theta)$ by making the denominator θ^n as *small* as possible.

3. **Apply the constraint:**
   Since θ cannot be smaller than the largest observed serial number, the smallest possible valid value for θ is $X_{(n)}$.

$$\hat{\theta}_{MLE} = X_{(n)} = \max(X_1, X_2, \ldots, X_n)$$

3. The returns on financial assets for (volatile) cryptocurrency are sometimes modeled using a Laplace distribution, which has heavier tails (extreme jumps are more common) than a Normal distribution.
   Suppose $X_1, \ldots, X_n$ are independent observations from a Laplace distribution with probability density function

$$f(x; \mu, b) = \frac{1}{2b} \exp\left(-\frac{|x - \mu|}{b}\right),$$

   where μ is an unknown location parameter and $b > 0$ is a known scale parameter.

   (a) Write down the likelihood and log-likelihood functions for the observed data $x_1, \ldots, x_n$.

   (b) Derive the maximum likelihood estimator of μ.

   (c) Compare this estimator with the MLE of the mean of a Normal distribution. Which estimator is less sensitive to extreme observations, and why?

   Solution:

   1. **Define the likelihood function:**

$$L(\mu) = \prod_{i=1}^{n} \frac{1}{2b} \exp\left(-\frac{|x_i - \mu|}{b}\right) = \left(\frac{1}{2b}\right)^n \exp\left(-\frac{1}{b}\sum_{i=1}^{n} |x_i - \mu|\right)$$

   2. **Take the log-likelihood:**

$$\ln L(\mu) = -n \ln(2b) - \frac{1}{b}\sum_{i=1}^{n} |x_i - \mu|$$

   3. **Identify the optimization objective:**
      To maximize $\ln L(\mu)$, we must maximize the negative term. Because $b > 0$, maximizing the log-likelihood is strictly equivalent to *minimizing* the sum of absolute deviations:

$$S(\mu) = \sum_{i=1}^{n} |x_i - \mu|$$

      (Note for students: This is L_1 loss, whereas Normal distribution MLE minimizes squared error, L_2 loss).

   4. **Differentiate:**
      Standard calculus requires a slight adjustment because the absolute value function is not differentiable exactly at $x_i = \mu$. We use the signum function, denoted as $\text{sgn}(x)$, which is 1 if $x > 0$, -1 if $x < 0$, and 0 if $x = 0$.

$$\frac{d}{d\mu}|x_i - \mu| = -\text{sgn}(x_i - \mu)$$

      Taking the derivative of our sum $S(\mu)$ with respect to μ:

$$\frac{dS(\mu)}{d\mu} = -\sum_{i=1}^{n} \text{sgn}(x_i - \mu)$$

5. **Set to zero and solve:**
   To minimize the sum, we set the derivative to 0:

$$\sum_{i=1}^{n} \operatorname{sgn}(x_i - \mu) = 0$$

   Let's analyze what this equation means. The sum of the signs will be zero only if the number of positive terms equals the number of negative terms.

   In other words, the number of data points where $x_i > \mu$ must exactly equal the number of data points where $x_i < \mu$.

6. **Conclusion:**
   The value μ that splits a dataset perfectly in half (50% of the data above it, 50% below it) is the median.

$$\hat{\mu}_{MLE} = \text{Sample Median}(X_1, \ldots, X_n)$$

For a Normal distribution, the MLE for the center is the sample *mean* (minimizing squared distances). For a Laplace distribution, the MLE for the center is the sample *median* (minimizing absolute distances). Because the Laplace distribution assumes extreme outliers are more likely, its MLE naturally yields a robust statistic (the median) that ignores the magnitude of those extreme outliers.

4. A sample of 20 cigarettes is tested to determine nicotine content, and the average value observed is 1.2 mg. Suppose that the population variance is not known in advance, and the sample variance is 0.04.

   Assuming a normal population, compute a 99% two-sided confidence interval for the mean nicotine content of a cigarette.

Solution:

Since the population variance is unknown, we use the Student t-distribution.

We are given

$$\overline{x} = 1.2, \qquad n = 20, \qquad s^2 = 0.04.$$

Thus,

$$s = 0.2 \text{ and } \operatorname{se}(\overline{x}) = s/\sqrt{20}.$$

For a 99% confidence interval,

$$\alpha = 0.01, \qquad \frac{\alpha}{2} = 0.005.$$

The number of degrees of freedom is

$$n - 1 = 19.$$

From the t-table,

$$t_{0.005,19} = 2.861.$$

Therefore, the confidence interval for the population mean is

$$\overline{x} \pm t_{0.005,19} \times \operatorname{se}(\overline{x})$$

Substituting the values,

$$\overline{x} - t_{0.005,19}\frac{s}{\sqrt{n}} = 1.2 - 2.861\frac{0.2}{\sqrt{20}} \approx 1.0721,$$
$$\overline{x} + t_{0.005,19}\frac{s}{\sqrt{n}} = 1.2 + 2.861\frac{0.2}{\sqrt{20}} \approx 1.3279.$$

Hence, the 99% two-sided confidence interval is

$$\boxed{(1.0721,\ 1.3279)}.$$

5. A random sample of 1200 engineers includes 300 Hispanic American engineers, 400 African American engineers. Assume no overlap.

   Determine 90% confidence intervals

   (a) For the proportion of all engineers who are Hispanic American or African American.

   (b) The excess in the proportion of African Americans and proportion of Hispanic Americans

Solution:

For a population proportion p, an approximate $100(1-\alpha)\%$ confidence interval is

$$\widehat{p} \pm z_{\alpha/2} \times \text{se}(\widehat{p}) = \widehat{p} \pm z_{\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}$$

provided the $np(1-p) > 10$ test is passed.

Here,

$$n = 1200, \qquad \alpha = 0.10, \qquad \frac{\alpha}{2} = 0.05.$$

From the standard normal table,

$$z_{0.05} = 1.645.$$

(a) The number of engineers who are Hispanic American or African American is

$$300 + 400 = 700$$

Thus,

$$\widehat{p} = \frac{700}{1200} \approx 0.58.$$

$1200\widehat{p}(1-\widehat{p}) = 292$ so the normal approximation check passes.

Therefore,

$$\widehat{p} \pm z_{0.05}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}} = 0.58 \pm 1.645\sqrt{\frac{(0.58)(0.42)}{1200}}.$$

Hence, the 90% confidence interval is approximately

$$\boxed{(0.56,\ 0.61)}.$$

(b) For the case of excess,

$$p_H = 300/1200 = 0.25 \text{ and } p_A = 400/1200 = 0.33333$$

Again $1200p_H(1-p_H)$ and $1200p_A * (1-p_A)$ exceeds 10, so the normal approximation check passes.

The difference is thus $0.33333 - 0.25 = 0.08333$ and the variance for the diffence is

$$[p_A + p_H - (p_A + p_H)^2]/n = (0.58333 - 0.006944)/1200 = 0.00048032$$

so the standard error is 0.021916

The confidence interval therefore is 1.645 times the standard error

$$\boxed{(0.047,\ 0.119)}.$$

6. It is claimed that a certain type of bipolar transistor has a mean current gain of at least 210. A sample of these transistors has sample mean current gain 200 and sample standard deviation 35.

Assuming that the population is normal, determine whether the claim should be rejected at the 5% level of significance if

(a) the sample size is 25;

(b) the sample size is 64.

Solution:

We test

$$H_0 : \mu \geq 210 \qquad \text{against} \qquad H_1 : \mu < 210.$$

Since the population variance is unknown, we use the statistic

$$T = \frac{\overline{X} - \mu_0}{S/\sqrt{n}}.$$

(a) For $n = 25$,

$$\overline{x} = 200, \qquad s = 35, \qquad \mu_0 = 210.$$

Thus,

$$\begin{aligned} t &= \frac{200 - 210}{35/\sqrt{25}} \\ &= \frac{-10}{7} \\ &\approx -1.429. \end{aligned}$$

The number of degrees of freedom is

$$25 - 1 = 24.$$

The p-value corresponding to the estimate is thus 0.08 which exceeds 0.05, so we fail to reject the claim at the 5% level of significance.

Equivalently, for a lower-tailed test at level $\alpha = 0.05$,

$$t_{0.05,24} \approx 1.711.$$

Hence the rejection region is

$$T < -1.711.$$

Since

$$-1.429 > -1.711,$$

we do not reject H_0.

Therefore,

$$\boxed{\text{The claim is not rejected when } n = 25.}$$

(b) For $n = 64$,

$$\begin{aligned} t &= \frac{200 - 210}{35/\sqrt{64}} \\ &= \frac{-10}{35/8} \\ &\approx -2.286. \end{aligned}$$

The number of degrees of freedom is

$$64 - 1 = 63.$$

The corresponding p-value is 0.013 which is less than 0.05. Therefore we reject the claim at the 5% signficance level.

Equivalently, for a lower-tailed test at level $\alpha = 0.05$,

$$t_{0.05,63} \approx 1.669.$$

Hence the rejection region is

$$T < -1.669.$$

Since

$$-2.286 < -1.669,$$

we reject H_0.

Therefore,

$$\boxed{\text{The claim is rejected when } n = 64.}$$

Thus, although the observed sample mean and sample standard deviation are the same in both cases, the larger sample size provides stronger evidence against the claim.

CS 215: Task 09: Due: 06-Nov; time as announced on Piazza.

- Your answers should be in your handwriting. Try to be neat and clear. If an LLM can read your handwriting, we might be able to.
- If you used any source (person or thing) explicitly state it. This is an individual task.
- For each of the two computer-based tasks, add to the zip package `a.pdf`, `a.py`, `b.pdf` and `b.py` as your solution(s).

---

1. Suppose the response follows the simple linear regression model

$$Y = \alpha + \beta x + e,$$

where x is fixed, $\mathbb{E}[e] = 0$, and $0 < \beta < 1$. Show that if

$$x < \frac{\alpha}{1-\beta},$$

then

$$x < \mathbb{E}[Y] < \frac{\alpha}{1-\beta}.$$

2. Consider the simple linear regression model

$$Y_i = \alpha + \beta x_i + e_i, \qquad i = 1, \ldots, n,$$

where the predictors are fixed and the errors are independent random variables with mean 0 and variance $\sigma^2 > 0$. Let A and B be the ordinary least squares estimators, and define

$$\widehat{Y}_i = A + Bx_i, \qquad R_i = Y_i - \widehat{Y}_i.$$

Let, as usual, $\bar{x} = n^{-1} \sum_i x_i$ and $S_{xx} = \sum_i (x_i - \bar{x})^2 > 0$.

(a) Derive $\mathrm{Cov}(A, B)$ in terms $\bar{x}$, σ^2 and S_{xx}. Recall that $\mathrm{Var}(B) = \frac{\sigma^2}{S_{xx}}$ as mentioned in class.

(b) Under what condition on the predictor values is $\mathrm{Cov}(A, B) = 0$?

(c) A student writes on Piazza: "The observation Y_i has variance σ^2. Subtracting the estimated fitted value $\widehat{Y}_i$ adds uncertainty" Dissect the student's argument explaining where the student is right. Is the intended conclusion "R_i must have variance at least σ^2" true or false?
Also explain intuitively your answer.

3. Suppose the observations follow the simple linear regression model

$$Y_i = \alpha + \beta x_i + e_i, \qquad i = 1, \ldots, n,$$

where the predictors are fixed, $S_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2 > 0$, and the errors are independent $N(0, \sigma^2)$ random variables. Consider the alternative parameterization

$$V_i = \gamma + \beta(x_i - \bar{x}) + e_i, \qquad \gamma = \alpha + \beta\bar{x}. \tag{1}$$

Let (A, B) and (C, B^*) denote the maximum likelihood estimators under the original and alternative parameterizations, respectively. Here C estimates γ, and B^* estimates β in Eq. 1. The coefficient estimators maximize the Gaussian likelihood with respect to the coefficients for each fixed $\sigma^2 > 0$.

(a) Reason the equivalence of V_i and Y_i and thus $B^* = B$. Also mathematically show this.

(b) Show that $C = \bar{Y}$ and derive its distribution.

4. The response random variables $Y_1, \ldots, Y_n$ follow the model

$$Y_i = \theta x_i^2 + e_i, \qquad i = 1, \ldots, n,$$

where $x_1, \ldots, x_n$ are fixed known constants, not all zero, and $e_1, \ldots, e_n$ are independent $N(0, \sigma^2)$ random variables, with known $\sigma^2 > 0$. The parameter $\theta \in \mathbb{R}$ is unknown.

(a) Find the least squares estimator of θ.

(b) Derive the maximum likelihood estimator of θ.

(c) Which estimator from that obtained in (a) and (b) is unbiased? Find its variance.

5. This is a computational exercise. Use Python, the numpy and matplot packages. We estimate about 10-20 lines of python code.

   Ten experiments on wood samples result in the following data. The input x_i is specific gravity and y_i is the observed maximum crushing strength parallel to the grain, measured in psi.

i	x_i	y_i	i	x_i	y_i
1	.41	1850	6	.39	1760
2	.46	2620	7	.41	2500
3	.44	2340	8	.44	2750
4	.47	2690	9	.43	2730
5	.42	2160	10	.44	3120

   Consider $Y_i = \alpha + \beta x_i + e_i$, where the inputs are fixed and the errors are independent $N(0, \sigma^2)$ random variables. We write A, B for the least squares estimators and a, b for their observed values.

   (a) Draw a scatter plot and fit the regression line. See the sample shown.

   (b) What is a, b and the estimated mean strength at $x = .43$ (Why this number?)

   (c) Estimate the variance.

   Use the fitted values a, b, s^2 as the fixed parameters of a simulation model. Perform exactly $M = 100$ independent repetitions of the ten-experiment design. Use the random number generator seed as 215 (Why?). Do not resample or change the input levels.

a-solution-plots.png

(d) Compute the empirical means and sample variances of the 100 intercept and slope estimates.

(e) Compare them with the theoretical means and variances under this simulation model.

(f) Plot the simulated slope estimates.

6. **Multiple regression: investigate the computation.** This is a computational exercise; Python packages may be used. Ten observations of four predictors and a response are provided below. Fit the model

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4} + e_i,$$

with fixed inputs and independent $N(0, \sigma^2)$ errors. Include an intercept. Denote the design matrix by X and the observed response vector by $\mathbf{y}$.

x_1	x_2	x_3	x_4	y
0	0	5	7	97
16	4	1	11	275
9	-3	8	4	246
4	2	3	9	140
25	-5	10	2	482
1	-1	6	6	122
9	3	2	10	183
16	-4	9	3	359
1	1	4	8	82
4	-2	7	5	146

(a) First attempt to compute the coefficient estimates by solving

$$(X^\mathsf{T} X)\mathbf{b} = X^\mathsf{T}\mathbf{y}$$

using `numpy.linalg.solve`. Record any error or warning. If values are returned, retain them for investigation rather than assuming success.

(b) Obtain an equivalent model with a full-column-rank matrix and fit it. Exhibit two distinct coefficient vectors for the original model that results in the same fitted responses.

CS 215: Task 09: Due: Nov-05; time as announced on Piazza.

1. Suppose the response follows the simple linear regression model

$$Y = \alpha + \beta x + e,$$

where x is fixed, $\mathbb{E}[e] = 0$, and $0 < \beta < 1$. Show that if

$$x < \frac{\alpha}{1-\beta},$$

then

$$x < \mathbb{E}[Y] < \frac{\alpha}{1-\beta}.$$

Solution: Since x, α, and β are fixed,

$$\mathbb{E}[Y] = \alpha + \beta x + \mathbb{E}[e] = \alpha + \beta x.$$

First, since $1 - \beta > 0$,

$$(1-\beta)x < \alpha.$$

Adding βx to both sides yields

$$x < \alpha + \beta x = \mathbb{E}[Y].$$

For the second inequality, multiply the assumption by $\beta(1-\beta) > 0$ to obtain

$$\beta(1-\beta)x < \beta(1-\beta)\frac{\alpha}{1-\beta} = \alpha\beta.$$

Adding $(1-\beta)\alpha$ to both sides gives

$$(1-\beta)(\alpha + \beta x) < \alpha\beta + (1-\beta)\alpha = \alpha.$$

Dividing by $1 - \beta > 0$,

$$\mathbb{E}[Y] = \alpha + \beta x < \frac{\alpha}{1-\beta}.$$

Combining the two inequalities,

$$\boxed{x < \mathbb{E}[Y] < \frac{\alpha}{1-\beta}.}$$

2. Consider the simple linear regression model

$$Y_i = \alpha + \beta x_i + e_i, \qquad i = 1, \ldots, n,$$

where the predictors are fixed and the errors are independent random variables with mean 0 and variance $\sigma^2 > 0$. Let A and B be the ordinary least squares estimators, and define

$$\widehat{Y}_i = A + Bx_i, \qquad R_i = Y_i - \widehat{Y}_i.$$

Let, as usual, $\bar{x} = n^{-1}\sum_i x_i$ and $S_{xx} = \sum_i (x_i - \bar{x})^2 > 0$.

(a) Derive $\mathrm{Cov}(A, B)$ in terms $\bar{x}$, σ^2 and S_{xx}. Recall that $\mathrm{Var}(B) = \frac{\sigma^2}{S_{xx}}$ as mentioned in class.

Solution: Recall the ordinary least squares estimators:

$$B = \frac{\sum_{i=1}^n (x_i - \bar{x})Y_i}{S_{xx}}, \qquad A = \bar{Y} - B\bar{x}.$$

Since $\bar{x}$ is fixed,

$$\begin{aligned}\mathrm{Cov}(A, B) &= \mathrm{Cov}(\bar{Y} - \bar{x}B, B) \\ &= \mathrm{Cov}(\bar{Y}, B) - \bar{x}\,\mathrm{Var}(B).\end{aligned}$$

We are given

$$\mathrm{Var}(B) = \frac{\sigma^2}{S_{xx}}.$$

Thus, it remains to calculate $\mathrm{Cov}(\bar{Y}, B)$.
Writing $d_i = x_i - \bar{x}$,

$$\bar{Y} = \frac{1}{n}\sum_{j=1}^n Y_j, \qquad B = \frac{1}{S_{xx}}\sum_{i=1}^n d_i Y_i.$$

Using bilinearity of covariance,

$$\mathrm{Cov}(\bar{Y}, B) = \frac{1}{nS_{xx}}\sum_{j=1}^n\sum_{i=1}^n d_i \,\mathrm{Cov}(Y_j, Y_i).$$

Because the predictors are fixed, the randomness in Y_i comes only from e_i. Independence of the errors gives

$$\mathrm{Cov}(Y_j, Y_i) = \begin{cases}\sigma^2, & i = j, \\ 0, & i \neq j.\end{cases}$$

Therefore, only terms with $i = j$ remain:

$$\begin{aligned}\mathrm{Cov}(\bar{Y}, B) &= \frac{\sigma^2}{nS_{xx}}\sum_{i=1}^n d_i \\ &= \frac{\sigma^2}{nS_{xx}}\sum_{i=1}^n (x_i - \bar{x}) \\ &= 0.\end{aligned}$$

The last equality follows from the defining property of the sample mean: $\sum_i (x_i - \bar{x}) = 0$.
Consequently,

$$\boxed{\mathrm{Cov}(A, B) = -\frac{\bar{x}\sigma^2}{S_{xx}}.}$$

Interpretation. Since $A = \bar{Y} - \bar{x}B$, an increase in the estimated slope must be accompanied by an adjustment in the intercept. The sign of this relationship depends on $\bar{x}$.

(b) Under what condition on the predictor values is $\mathrm{Cov}(A, B) = 0$?

Solution: From part (a),

$$\mathrm{Cov}(A, B) = -\frac{\bar{x}\sigma^2}{S_{xx}}.$$

Since $\sigma^2 > 0$ and $S_{xx} > 0$, the covariance vanishes if and only if

$$\boxed{\bar{x} = 0 \iff \sum_{i=1}^{n} x_i = 0.}$$

(c) A student writes on Piazza: "The observation Y_i has variance σ^2. Subtracting the estimated fitted value $\widehat{Y}_i$ adds uncertainty" Dissect the student's argument explaining where the student is right. Is the intended conclusion "R_i must have variance at least σ^2" true or false?

Also explain intuitively your answer.

Solution: The student's observation that $\widehat{Y}_i$ is random is correct: the fitted value depends on the estimated coefficients A and B.

However, the conclusion is **false**.

The key mistake is implicitly treating Y_i and $\widehat{Y}_i$ as independent or uncorrelated.

In general,

$$\mathrm{Var}(Y_i - \widehat{Y}_i) = \mathrm{Var}(Y_i) + \mathrm{Var}(\widehat{Y}_i) - 2\,\mathrm{Cov}(Y_i, \widehat{Y}_i).$$

The student correctly identifies the first two terms, but ignores the covariance term.

Intuition. The fitted value $\widehat{Y}_i$ is not an independent noisy quantity: it is constructed using the same observations, including Y_i.

In fact, it can be shown that

$$\mathrm{Var}(R_i) = \sigma^2 \left(1 - \frac{1}{n} - \frac{(x_i - \bar{x})^2}{S_{xx}}\right).$$

so the variance is actually smaller in general.

3. Suppose the observations follow the simple linear regression model

$$Y_i = \alpha + \beta x_i + e_i, \qquad i = 1, \ldots, n,$$

where the predictors are fixed, $S_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2 > 0$, and the errors are independent $N(0, \sigma^2)$ random variables. Consider the alternative parameterization

$$V_i = \gamma + \beta(x_i - \bar{x}) + e_i, \qquad \gamma = \alpha + \beta\bar{x}. \tag{1}$$

Let (A, B) and (C, B^*) denote the maximum likelihood estimators under the original and alternative parameterizations, respectively. Here C estimates γ, and B^* estimates β in Eq. 1. The coefficient estimators maximize the Gaussian likelihood with respect to the coefficients for each fixed $\sigma^2 > 0$.

(a) Reason the equivalence of V_i and Y_i and thus $B^* = B$. Also mathematically show this.

(b) Show that $C = \bar{Y}$ and derive its distribution.

Solution:

(a) **Equivalence and equality of slope estimators.**

The alternative model is obtained by centering the predictor around its sample mean.

Indeed, using $\gamma = \alpha + \beta\bar{x}$,

$$\begin{aligned} V_i &= \gamma + \beta(x_i - \bar{x}) + e_i \\ &= \alpha + \beta\bar{x} + \beta x_i - \beta\bar{x} + e_i \\ &= \alpha + \beta x_i + e_i \\ &= Y_i. \end{aligned}$$

Thus, the two parameterizations describe exactly the same observations and fitted regression line.

Only the interpretation of the intercept changes: α is the fitted mean at $x = 0$, while γ is the fitted mean at $x = \bar{x}$.

Since shifting the origin of the predictor does not change its relationship with the response, the estimated slope should remain unchanged.

To show this mathematically, let

$$z_i = x_i - \bar{x}.$$

Then $\bar{z} = 0$, and the centered-model residual sum of squares is

$$Q(\gamma, \beta) = \sum_{i=1}^{n}(Y_i - \gamma - \beta z_i)^2.$$

Differentiating with respect to β,

$$\frac{\partial Q}{\partial \beta} = -2\sum_{i=1}^{n} z_i(Y_i - \gamma - \beta z_i).$$

At the minimum,

$$\sum_i z_i Y_i - \gamma \sum_i z_i - \beta \sum_i z_i^2 = 0.$$

Because $\sum_i z_i = 0$,

$$B^* = \frac{\sum_i z_i Y_i}{\sum_i z_i^2} = \frac{\sum_i (x_i - \bar{x}) Y_i}{S_{xx}}.$$

For the original model, the OLS slope estimator is

$$B = \frac{\sum_i (x_i - \bar{x})(Y_i - \bar{Y})}{S_{xx}}.$$

But

$$\sum_i (x_i - \bar{x})\bar{Y} = \bar{Y} \sum_i (x_i - \bar{x}) = 0.$$

Therefore,

$$\boxed{B^* = B.}$$

(b) **Estimator and distribution of C.**

To find the intercept estimator, differentiate the centered-model residual sum of squares with respect to γ:

$$\frac{\partial Q}{\partial \gamma} = -2 \sum_{i=1}^{n} (Y_i - \gamma - \beta z_i).$$

Setting this derivative equal to zero at the optimum,

$$nC = \sum_{i=1}^{n} Y_i - B^* \sum_{i=1}^{n} z_i.$$

Since the centered predictors sum to zero,

$$\boxed{C = \bar{Y}.}$$

To derive its distribution, average the original model:

$$\begin{aligned} \bar{Y} &= \frac{1}{n} \sum_{i=1}^{n} (\alpha + \beta x_i + e_i) \\ &= \alpha + \beta \bar{x} + \bar{e} \\ &= \gamma + \bar{e}, \end{aligned}$$

where

$$\bar{e} = \frac{1}{n} \sum_{i=1}^{n} e_i.$$

Since the errors are independent normal random variables, their average is also normally distributed, with

$$\mathbb{E}[\bar{e}] = 0, \qquad \mathrm{Var}(\bar{e}) = \frac{1}{n^2} \sum_{i=1}^{n} \sigma^2 = \frac{\sigma^2}{n}.$$

Therefore,

$$\boxed{C = \bar{Y} \sim N\left(\gamma, \frac{\sigma^2}{n}\right).}$$

4. The response random variables $Y_1, \ldots, Y_n$ follow the model

$$Y_i = \theta x_i^2 + e_i, \qquad i = 1, \ldots, n,$$

where $x_1, \ldots, x_n$ are fixed known constants, not all zero, and $e_1, \ldots, e_n$ are independent $N(0, \sigma^2)$ random variables, with known $\sigma^2 > 0$. The parameter $\theta \in \mathbb{R}$ is unknown.

(a) Find the least squares estimator of θ.

(b) Derive the maximum likelihood estimator of θ.

(c) Which estimator from that obtained in (a) and (b) is unbiased? Find its variance.

Solution:

(a) **Least squares.** For observed responses $y_1, \ldots, y_n$, minimize

$$Q(\theta) = \sum_{i=1}^{n} (y_i - \theta x_i^2)^2.$$

Differentiating gives

$$Q'(\theta) = -2 \sum_i x_i^2 y_i + 2\theta D, \qquad Q''(\theta) = 2D > 0.$$

where $D = \sum_{i=1}^{n} x_i^4$ assumed to be not equal to 0.

Thus the unique least squares estimate and its corresponding estimator are

$$\boxed{t_{\mathrm{LS}} = \frac{\sum_i x_i^2 y_i}{D}, \qquad T_{\mathrm{LS}} = \frac{\sum_i x_i^2 Y_i}{D}.}$$

(b) **Maximum likelihood.**

With the errors are independent normal random variables with mean zero and known variance $\sigma^2 > 0$, the observed-data likelihood as a function of θ is (the notation for the likelihood is slightly different from the earlier task)

$$L(\theta; y_1, \ldots, y_n) = (2\pi\sigma^2)^{-n/2} \exp\left\{ -\frac{1}{2\sigma^2} \sum_i (y_i - \theta x_i^2)^2 \right\}.$$

Its logarithm is

$$\ell(\theta; y_1, \ldots, y_n) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{Q(\theta)}{2\sigma^2}.$$

Maximizing this is equivalent to minimizing Q. Consequently,

$$t_{\mathrm{ML}} = t_{\mathrm{LS}}, \qquad \boxed{T_{\mathrm{ML}} = T_{\mathrm{LS}} = \frac{\sum_i x_i^2 Y_i}{D}.}$$

(c) **Expectation and variance.** Denote the common estimator by T. Substitution of the model gives

$$T = \frac{\sum_i x_i^2 (\theta x_i^2 + e_i)}{D} = \theta + \frac{\sum_i x_i^2 e_i}{D}.$$

Hence

$$\mathbb{E}_\theta[T] = \theta, \qquad \mathrm{Var}_\theta(T) = \frac{\sigma^2 \sum_i x_i^4}{D^2} = \frac{\sigma^2}{D}.$$

Therefore the common estimator is unbiased, with

$$\boxed{T = \frac{\sum_i x_i^2 Y_i}{\sum_i x_i^4}, \qquad \mathrm{Var}_\theta(T) = \frac{\sigma^2}{\sum_i x_i^4}.}$$

Interpretation. Defining $z_i = x_i^2$ implies $Y_i = \theta z_i + e_i$, a straight-line regression through the origin in the transformed predictor z_i. The model is linear in the unknown parameter θ, even though the mean response is quadratic in the original input x_i.

5. This is a computational exercise. Use Python, the numpy and matplot packages. We estimate about 10-20 lines of python code.

   Ten experiments on wood samples result in the following data. The input x_i is specific gravity and y_i is the observed maximum crushing strength parallel to the grain, measured in psi.

i	x_i	y_i	i	x_i	y_i
1	.41	1850	6	.39	1760
2	.46	2620	7	.41	2500
3	.44	2340	8	.44	2750
4	.47	2690	9	.43	2730
5	.42	2160	10	.44	3120

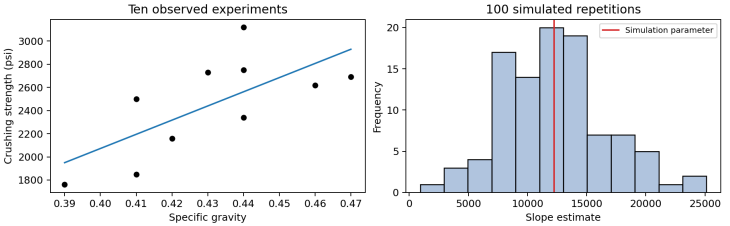
   Consider $Y_i = \alpha + \beta x_i + e_i$, where the inputs are fixed and the errors are independent $N(0, \sigma^2)$ random variables. We write A, B for the least squares estimators and a, b for their observed values.

   (a) Draw a scatter plot and fit the regression line. See the sample shown.

   (b) What is a, b and the estimated mean strength at $x = .43$ (Why this number?)

   (c) Estimate the variance.

   Use the fitted values a, b, s^2 as the fixed parameters of a simulation model. Perform exactly $M = 100$ independent repetitions of the ten-experiment design. Use the random number generator seed as 215 (Why?). Do not resample or change the input levels.



   (d) Compute the empirical means and sample variances of the 100 intercept and slope estimates.

   (e) Compare them with the theoretical means and variances under this simulation model.

   (f) Plot the simulated slope estimates.

Solution:

(a) With $n = 10$, the summary quantities are

$$\bar{x} = .431, \qquad \bar{y} = 2452, \qquad S_{xx} = .00529.$$

Estimators are $B = S_{xY}/S_{xx}$ and $A = \bar{Y} - B\bar{x}$. Their observed values are

$$b \approx 12245.75, \qquad a \approx -2825.92, \qquad \widehat{m}(.43) = a + .43b \approx 2439.75 \text{ psi}.$$

(b) The residuals, in table order $i = 1, \ldots, 10$, are approximately

$$(r_1, \ldots, r_5) \approx (-344.839, -187.127, -222.212, -239.584, -157.297),$$
$$(r_6, \ldots, r_{10}) \approx (-189.924, 305.161, 187.788, 290.246, 557.788).$$

Direct computation results in

$$\text{ssr} \approx 845280.5293 \text{ psi}^2, \qquad s^2 \approx 105660.07 \text{ psi}^2.$$

Here s^2 is an observed value of the unbiased estimator $S^2 = \text{SSR}/(n-2)$ of the common error variance σ^2.

(c) Denote the simulation parameters by $\alpha_0 = a$, $\beta_0 = b$, and $\sigma_0^2 = s^2$. These are fixed numbers in the simulation, not the unknown true parameters of the original population.

In each repetition the estimators satisfy

$$\mathbb{E}[A] = \alpha_0 \approx -2825.92, \quad \text{Var}(A) = \sigma_0^2 \left(\frac{1}{10} + \frac{\bar{x}^2}{S_{xx}} \right),$$

$$\mathbb{E}[B] = \beta_0 \approx 12245.75, \quad \text{Var}(B) = \frac{\sigma_0^2}{S_{xx}}.$$

The theoretical variances are approximately 3.72×10^6 and 1.99×10^7, respectively.

With the code below the empirical means are approximately -2891.90 and 12388.95, and the empirical variances are 3.58×10^6 and 1.93×10^7.

The following code implements the experiment;

`ddof=1` gives the sample variances across repetitions.

```
import numpy as np
import matplotlib.pyplot as plt
x = np.array([.41,.46,.44,.47,.42,.39,.41,.44,.43,.44])
y = np.array([1850,2620,2340,2690,2160,1760,2500,2750,2730,3120])
X = np.column_stack((np.ones(x.size), x))
a, b = np.linalg.lstsq(X, y, rcond=None)[0]
r = y - X @ np.array([a, b])
s2 = (r @ r) / (x.size - 2)
rng = np.random.default_rng(215)
ys = a + b*x[:, None] + rng.normal(0, np.sqrt(s2), (10, 100))
coef = np.linalg.lstsq(X, ys, rcond=None)[0]
print(coef.mean(axis=1))
print(coef.var(axis=1, ddof=1))
plt.hist(coef[1], bins=12, edgecolor="black")
plt.axvline(b, color="red", label="Simulation slope parameter")
plt.xlabel("Observed slope estimate in one repetition")
plt.ylabel("Frequency")
plt.legend()
plt.show()
```

6. **Multiple regression: investigate the computation.** This is a computational exercise; Python packages may be used. Ten observations of four predictors and a response are provided below. Fit the model

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4} + e_i,$$

with fixed inputs and independent $N(0, \sigma^2)$ errors. Include an intercept. Denote the design matrix by X and the observed response vector by $\mathbf{y}$.

x_1	x_2	x_3	x_4	y
0	0	5	7	97
16	4	1	11	275
9	-3	8	4	246
4	2	3	9	140
25	-5	10	2	482
1	-1	6	6	122
9	3	2	10	183
16	-4	9	3	359
1	1	4	8	82
4	-2	7	5	146

(a) First attempt to compute the coefficient estimates by solving

$$(X^\mathsf{T} X)\mathbf{b} = X^\mathsf{T}\mathbf{y}$$

using `numpy.linalg.solve`. Record any error or warning. If values are returned, retain them for investigation rather than assuming success.

(b) Obtain an equivalent model with a full-column-rank matrix and fit it. Exhibit two distinct coefficient vectors for the original model that results in the same fitted responses.

Solution:

(a) In the tested NumPy environment, solving the normal equations raises a `LinAlgError` with message "Singular matrix."

(b) For every observation,

$$x_{i2} + x_{i3} = 5, \qquad x_{i3} + x_{i4} = 12.$$

Consequently, the five columns of X span only the three-dimensional space generated by $\mathbf{1}$, the x_1 column, and the x_3 column. These three columns are independent, so $\operatorname{rank}(X) = 3$. Thus $X^\mathsf{T} X$ is singular and the original coefficient vector is not uniquely determined. A pseudoinverse-based solver such as `numpy.linalg.lstsq` can select one solution (the minimum Euclidean norm solution), but it cannot make the original parameters identifiable.

Multiple solutions are therefore possible. Here is one.

Substituting $x_2 = 5 - x_3$ and $x_4 = 12 - x_3$ results in

$$\mathbb{E}[Y_i] = \gamma_0 + \gamma_1 x_{i1} + \gamma_3 x_{i3},$$

where

$$\gamma_0 = \beta_0 + 5\beta_2 + 12\beta_4, \qquad \gamma_1 = \beta_1, \qquad \gamma_3 = \beta_3 - \beta_2 - \beta_4.$$

These three combinations are identifiable. Their observed least squares estimates result in

$$\widehat{m}(x_1, x_3) = 43.32 + 13.94_1 + 9.36_3.$$

In the order $(\beta_0, \beta_1, \beta_2, \beta_3, \beta_4)$, two possible coefficient estimates are

$$\mathbf{b}^{(1)} = (43.32,\ 13.94,\ 0,\ 9.36,\ 0)^{\mathsf{T}},$$
$$\mathbf{b}^{(2)} = (38.32,\ 13.94,\ 1,\ 10.35,\ 0)^{\mathsf{T}}.$$